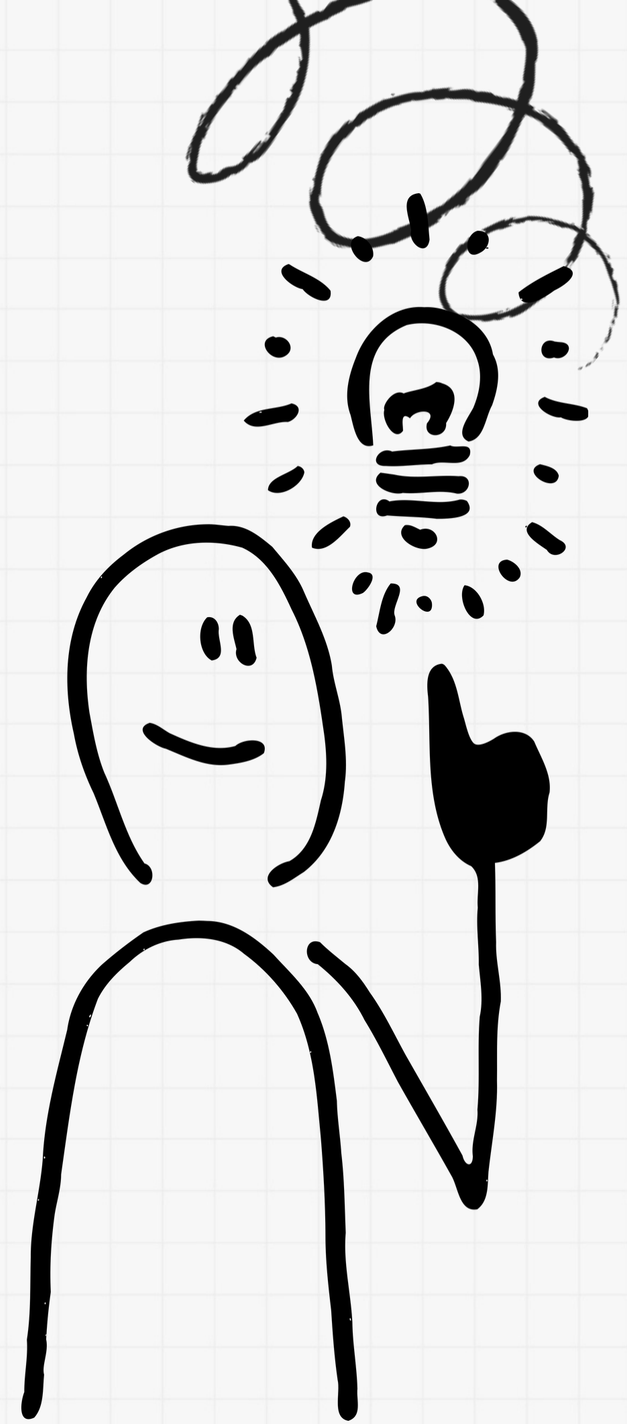
Self-RAG

Learning to Retrieve, Generate, and Critique through Self-Reflection

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CS613 · Paper Presentation

Presented by: [Team member names]

Roadmap

1 · Background & problem

LMs as distributions, retrieval, RAG, why plain RAG fails, prior work

2 · Core idea

Reflection tokens, what is new, comparison with RAG variants

3 · Method

Formalisation, critic + generator training, inference maths, cost, why it works

4 · Experiments

Tasks, metrics, baselines, full results, ablations, analyses

5 · Critical analysis

Strengths, limitations, is the evidence sufficient?, improvements

6 · Impact

Follow-up work and modern LLM systems

7 · Wrap-up

Takeaways, team contribution, references, Q&A

Backup

Q&A bank, GPT-4 prompts, evaluation details

01

Background & problem

- Language models as probability distributions
- Parametric vs non-parametric knowledge
- Retrieval: sparse and dense
- RAG formulations and their failure modes
- Prior work and the gap Self-RAG fills



Language models are probability distributions

Chain rule (exact)

$$p_\theta(w_{1:n}) = \prod_{i=1}^{n} p_\theta(w_i \mid w_{<i})$$

Softmax over the vocabulary

$$p_\theta(w_i = v \mid w_{<i}) = \frac{\exp(z_{i,v})}{\sum_{u \in V} \exp(z_{i,u})}, \qquad z_i = W h_i$$

Training: masked NLL

$$\mathcal{L}(\theta) = -\,\mathbb{E}_{(x,y)} \sum_i m_i \log p_\theta(y_i \mid x, y_{<i})$$

m_i (0 or 1) chooses which positions are trained. Instruction tuning masks the prompt; Self-RAG also masks retrieved passages.

- **Teacher forcing:** the model always conditions on the gold prefix, so one forward pass gives the loss at every position.
- **Pre-training:** NLL on web text gives a text continuer (Llama2-7B).
- **Instruction tuning (SFT):** NLL on (instruction, response) pairs gives an assistant (Alpaca). Self-RAG's generator is plain SFT on a special corpus.
- **RLHF:** train a reward model r, then maximise E[r] − β · KL(π || π_ref) with PPO (ChatGPT, Llama2-chat). Self-RAG borrows the learned critic but avoids RL.
- **Prompt as prefix:** conditional generation p(y | x) just treats x as the first part of the sequence.

Where does an LM keep its knowledge?

- **Parametric memory:** facts stored implicitly in θ. Fixed after training, no source, costly to update.
- **Non-parametric memory:** an external corpus (e.g. Wikipedia split into ~100-word passages). Updatable, inspectable, citable.
- **Hallucination:** fluent, confident, false output. The model samples what is plausible, not what is true.
- **Long tail:** rare entities appear in few training examples, so probability spreads over plausible wrong answers (Mallen et al. 2023).

no retrieval

with retrieval

14.7% ⇨ **45.7%**

Llama2-13B on PopQA long-tail (1,399 questions about entities with <100 monthly Wikipedia views). Paper Table 2.



Retrieval matters most exactly where the model is weakest.



Retrieval: sparse vs dense

Sparse: BM25

$$\mathrm{BM25}(q,d) = \sum_{t \in q} \mathrm{IDF}(t)\, \frac{f(t,d)\,(k_1+1)}{f(t,d) + k_1\left(1 - b + b\,\frac{|d|}{\mathrm{avgdl}}\right)}$$

$$\mathrm{IDF}(t) = \log \frac{N - n_t + 0.5}{n_t + 0.5}$$

- Exact word overlap; rare terms count more
- k1 about 1.2 (saturation), b about 0.75 (length normalisation)
- Misses synonyms and paraphrases

Dense: bi-encoder (Contriever)

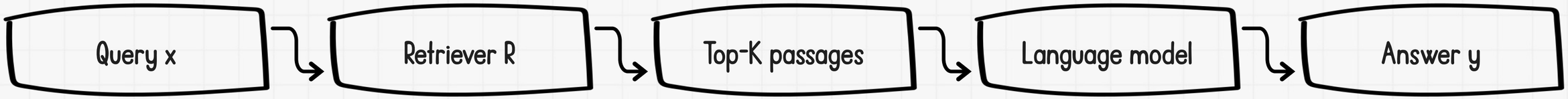
$$\mathrm{sim}(q,d) = \langle E(q),\ E(d) \rangle$$

$$\mathcal{L}_{\mathrm{InfoNCE}} = -\log \frac{e^{\mathrm{sim}(q,\,d^+)/\tau}}{e^{\mathrm{sim}(q,\,d^+)/\tau} + \sum_{j=1}^{m} e^{\mathrm{sim}(q,\,d_j^-)/\tau}}$$

- Passages embedded offline; search = maximum inner product search (FAISS) over ~21M passages
- Contriever: contrastive pre-training on random crops (no labels), then MS MARCO fine-tuning
- Self-RAG uses Contriever-MS MARCO, top-5 (up to 10 in training)

Key fact: Self-RAG never trains the retriever. R is frozen and off-the-shelf; all judgment of relevance happens inside the LM, after retrieval.

Retrieval-augmented generation: three formulations



$p_\eta(z \mid x) \propto \exp\langle E(z), E(x)\rangle$ is the retriever's distribution over passages; p_θ is the generator.

RAG-Sequence (Lewis 2020)

$$p(y|x) \approx \sum_{z \in \text{top}K} p_\eta(z|x) \prod_i p_\theta(y_i|x, z, y_{<i})$$

One passage explains the whole answer; marginalise over passages weighted by the retriever. Trained end-to-end.

RAG-Token (Lewis 2020)

$$p(y|x) \approx \prod_i \sum_{z \in \text{top}K} p_\eta(z|x)\, p_\theta(y_i|x, z, y_{<i})$$

Marginalise per token: different tokens can draw on different passages.

In-context RALM (Ram 2023)

$$y \sim p_\theta(\ \cdot\ \mid [d_1; \ldots; d_K;\ x])$$

Concatenate the K passages before the prompt. No training. Used by the Ret-ChatGPT / Ret-Llama2-chat baselines.

Why plain RAG falls short



Always retrieves

Top-K passages for every prompt, even 'write an essay about your summer'. Wastes compute and derails creative or reasoning tasks.

Ret-ChatGPT on PubHealth: 70.1 → 54.7



Irrelevant passages

The retriever's top hit can be off-topic; LMs are easily distracted by irrelevant context (Shi et al. 2023).

Alpaca-13B on PubHealth: 55.5 → 51.1 with retrieval



No grounding check

Nothing makes the output follow the passages and nothing checks it afterwards; models are not trained to use the evidence.

Open RAG models on ASQA: 2–8% citation precision

Formally: retrieval helps only if E[quality | retrieve] > E[quality | no retrieve], and that depends on x and on what has been written so far (y<t). A fixed 'always retrieve K' policy cannot capture this.

Prior work (1/2): when and what to retrieve

Method	Idea	Gap that Self-RAG targets
FLARE / Active RAG (Jiang 2023)	Retrieve when the next sentence has low-probability tokens; prompts a proprietary LLM	Heuristic trigger; no relevance or support judgment
Toolformer (Schick 2023)	Self-supervised learning of API calls (e.g. Wikipedia search)	Learns to call tools, not to judge what comes back
SAIL (Luo 2023)	Instruction-tune with top retrieved passages prepended	Always retrieves; indiscriminate
RA-DIT (Lin 2023)	Dual instruction tuning of retriever and LM	Always retrieves; no self-critique
RECOMP (Xu 2023), NLI filter (Yoran 2023)	Compress or filter passages with an extra model	Extra model at inference; relevance only, not support
Atlas (Izacard 2022b)	Retriever and LM pre-trained jointly, then few-shot fine-tuned	Task-specific; fixed retrieval policy

Costs reported for such methods: runtime efficiency (Mallen 2023), robustness to irrelevant context (Shi 2023), lack of attribution (Liu 2023a; Gao 2023).

Prior work (2/2): critics, control tokens, refinement

Method	Idea	Difference from Self-RAG
RLHF with PPO (Ouyang 2022)	Reward model from preferences, then PPO	Expensive online RL; behaviour fixed at training time
Fine-grained RLHF (Wu 2023)	Several fine-grained reward models	Still RL; no retrieval
CTRL, QUARK, Korbak 2023	Control tokens that condition generation	Tokens steer the output; they do not evaluate it
Self-evaluation decoding (Xie 2023)	Value-guided decoding for reasoning	One evaluation dimension; no retrieval
Self-Refine, CoVE, REFINER	Generate, give feedback, revise, by prompting	Many inference calls per answer
LATS (Zhou 2023)	Tree search guided by LM value scores	Single overall score; prompting only

Self-RAG = adaptive retrieval + learned, fine-grained critique + controllable decoding, trained with plain SFT.

What the authors set out to achieve



Retrieve only when useful

Decide per segment, from the input and what has been written



Judge every passage

Say whether each retrieved passage is relevant



Verify and cite

Check that each sentence is supported by its passage



Keep versatility

Creative and reasoning prompts behave like a normal LM



Control at test time

Trade precision, fluency and cost without retraining



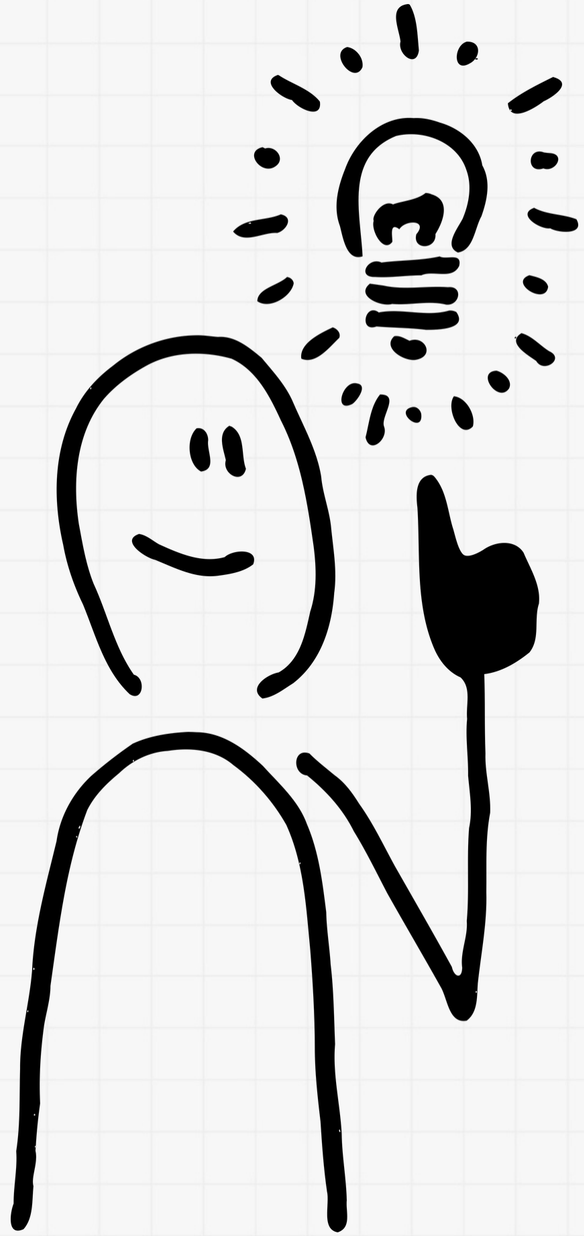
Train cheaply

One model, standard next-token loss, no RL

02

Core idea

- Reflection tokens: retrieval and critique as vocabulary
- Conventional RAG vs Self-RAG, step by step
- Exact token definitions
- What is new



Conventional RAG vs Self-RAG (paper Fig. 1)

Conventional RAG

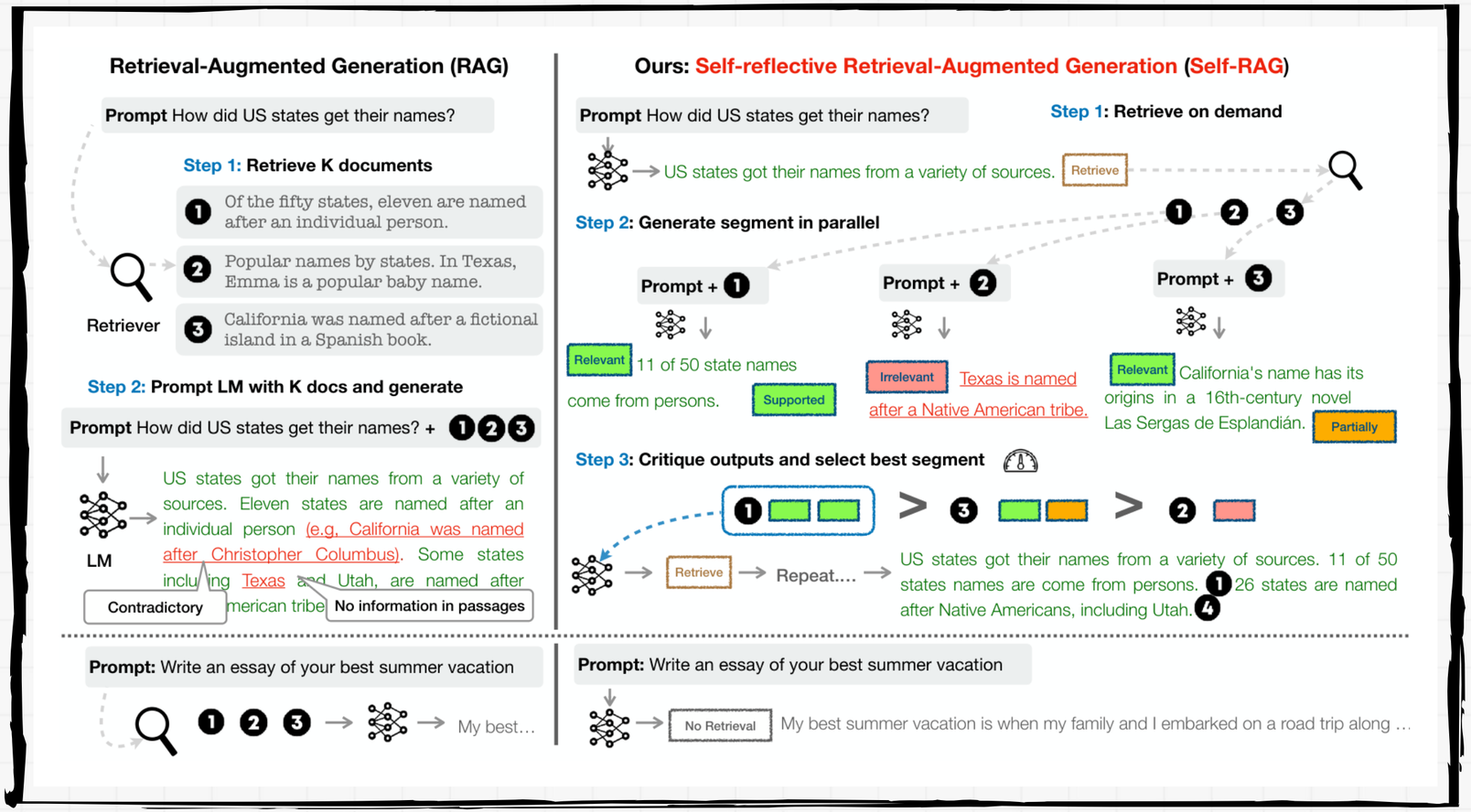
- Step 1: retrieve K documents for the prompt, always
- Step 2: prompt the LM with all K documents and generate
- Q: How did US states get their names?
  - Output mixes in an error from an off-topic passage: 'California was named after Christopher Columbus'
  - Nothing re-checks the output against the passages
- Q: Write an essay about your best summer vacation
  - Still retrieves, although no facts are needed

Self-RAG

- Step 1, retrieve on demand: the model emits [Retrieve] = Yes
- Step 2, generate in parallel: one continuation per passage
  - d1 '11 of 50 state names come from persons': Relevant, Supported
  - d2 'Popular baby names in Texas': Irrelevant
  - d3 'California's name has its origins in a 16th-century novel': Relevant, Partially
- Step 3, critique and select: keep the best segment, then repeat
- Essay prompt: [Retrieve] = No, so it writes like a normal LM

From the paper: Figure 1

Asai et al. 2024, Fig. 1



- **Left (RAG):** always retrieves K = 3 documents and generates once; the answer contradicts a passage and the essay prompt still retrieves.
- **Right (Self-RAG):** Retrieve → one candidate per passage tagged Relevant / Irrelevant and Supported / Partially → ranking 1 > 3 > 2 → repeat; the essay prompt gets No Retrieval.

Reflection tokens (paper Table 1)

Add special tokens to the vocabulary and train ONE model, with ordinary next-token prediction, to write them between its own sentences. **Critique becomes next-token prediction.**

Retrieve

Should I fetch evidence for the next segment?

Yes · No · Continue

input: x, y<t
before each segment

IsRel

Does passage d give useful information for x?

Relevant (best) · Irrelevant

input: x, d
after the passage

IsSup

Is every verification-worthy claim in yt backed by d?

Fully (best) · Partially · No support

input: x, d, yt
after the segment

IsUse

Is y a useful response to x, regardless of truth?

5 (best) · 4 · 3 · 2 · 1

input: x, y
once, at the end

From the paper: Table 1 and Algorithm 1

Asai et al. 2024, Table 1 / Alg. 1

Type	Input	Output	Definitions
Retrieve	x / x, y	{yes, no, continue}	Decides when to retrieve with $\mathcal{R}$
ISREL	x, d	{**relevant**, irrelevant}	d provides useful information to solve x.
ISSUP	x, d, y	{**fully supported**, partially supported, no support}	All of the verification-worthy statement in y is supported by d.
ISUSE	x, y	{**5**, 4, 3, 2, 1}	y is a useful response to x.

Algorithm 1 SELF-RAG Inference

Require: Generator LM $\mathcal{M}$, Retriever $\mathcal{R}$, Large-scale passage collections $\{d_1, \ldots, d_N\}$

Input: input prompt x and preceding generation $y_{<t}$, **Output:** next output segment y_t
$\mathcal{M}$ predicts Retrieve given $(x, y_{<t})$
if Retrieve == `Yes` **then**
  Retrieve relevant text passages $\mathbf{D}$ using $\mathcal{R}$ given (x, y_{t-1}) ▷ Retrieve
  $\mathcal{M}$ predicts ISREL given x, d and y_t given $x, d, y_{<t}$ for each $d \in \mathbf{D}$ ▷ Generate
  $\mathcal{M}$ predicts ISSUP and ISUSE given x, y_t, d for each $d \in \mathbf{D}$ ▷ Critique
  Rank y_t based on ISREL , ISSUP , ISUSE ▷ Detailed in Section 3.3
else if Retrieve == `No` **then**
  $\mathcal{M}_{gen}$ predicts y_t given x ▷ Generate
  $\mathcal{M}_{gen}$ predicts ISUSE given x, y_t ▷ Critique

Token definitions in detail (App. A.1)

Retrieve

Yes: the continuation needs factual grounding. No: it does not, or retrieval would not help. Continue: keep using the passage already retrieved (e.g. several sentences from one Wikipedia intro).

IsRel

Relevant if the evidence provides useful information to solve x; otherwise Irrelevant.

IsSup

Attribution: how much of the output is entailed by the evidence (Menick 2022; Bohnet 2022). Three levels following Yue 2023 and Nakano 2021: Fully supported, Partially supported, No support / Contradictory.

IsUse

Perceived utility (Liu 2023a): is the response a helpful, informative answer, independent of whether it is factual. Also called plausibility (Menick 2022). Five-point scale, 1 lowest.

What is new

1

Retrieval on demand, per segment

The LM itself decides, mid-generation, using a learned token, not a confidence heuristic.



2

Self-critique as vocabulary

Relevance, support and utility are generated tokens, so judging costs one decoding step and needs no reward model at inference.

3

Critique learned offline, without RL

A critic labels the corpus once; the generator is trained with plain SFT.

4

Controllable decoding

Critique probabilities define a tunable score (weights w, threshold δ): one model, many behaviours, no retraining.

5

Fine-grained attribution

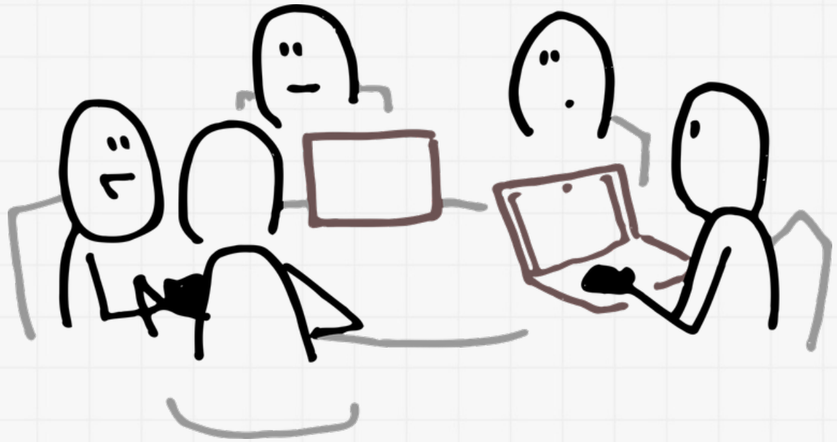
Every segment is generated from exactly one passage and carries its own support label: a built-in citation.

Self-RAG vs other RAG designs

	Lewis-RAG	In-context RAG	Self-RAG
When to retrieve	Always, once	Always, once	Model decides, per segment
How passages are combined	Soft mixture (marginalise)	Concatenate all K	One passage per candidate, then select
Relevance check	Retriever score only	None	LM predicts IsRel
Grounding check	None	None	LM predicts IsSup
Citations	No	Depends on prompt	Built in: each segment tied to one d
Test-time control	No	No	Yes: weights w and threshold δ
Training	End-to-end (retriever + LM)	None	SFT of the LM only; retriever frozen

03

Method



- Formalisation and token order
- Training: GPT-4 → critic → annotated corpus → generator
- Inference: threshold, critique scores, segment beam search
- Worked example, cost, why it works

Problem formalisation (§3.1)

Input x; output y = [y1, ..., yT] of T segments (one sentence each, split with spaCy). Each yt may contain words from V and reflection tokens from R; the model's vocabulary is V plus R.

Generative story of segment t

$$r_t^{ret} \sim p_{\mathcal{M}}(\cdot \mid x,\ y_{<t})$$

$$\text{if } r_t^{ret} = \text{Yes}: \quad d_t \leftarrow \mathcal{R}(x,\ y_{t-1}) \qquad \text{(external call)}$$

$$r_t^{rel} \sim p_{\mathcal{M}}(\cdot \mid x,\ d_t,\ y_{<t})$$

$$y_t \sim p_{\mathcal{M}}(\cdot \mid x,\ d_t,\ r_t^{rel},\ y_{<t})$$

$$r_t^{sup} \sim p_{\mathcal{M}}(\cdot \mid x,\ d_t,\ y_t, \ldots)$$

$$\text{else}: \quad y_t \sim p_{\mathcal{M}}(\cdot \mid x,\ y_{<t})$$

$$\text{after } y_T: \quad r^{use} \sim p_{\mathcal{M}}(\cdot \mid x,\ y)$$

Notes

- The retriever query at step t is (x, y_{t-1})
- Passages d are inputs supplied by R, not tokens produced by M
- Segment unit is a design choice; the framework also allows sub-sentences
- One model plays three roles: retrieval policy, writer, critic

One token stream, so the chain rule applies

$$p_{\mathcal{M}}(\tilde{y} \setminus D \mid x) = \prod_{t=1}^{T} p(r_t^{ret} \mid \cdot)\, \left[p(r_t^{rel} \mid \cdot)\, p(y_t \mid \cdot)\, p(r_t^{sup} \mid \cdot) \right]^{\mathbf{1}[ret_t]} p(y_t \mid \cdot)^{1-\mathbf{1}[ret_t]}\, p(r^{use} \mid \cdot)$$

ỹ is the interleaved sequence; D (retrieved passages) is excluded from the product because R, not M, produces it. Each '·' means 'everything before it in the stream'.

Why this token order?

Retrieve before the segment — you must decide to look things up before writing

IsRel after the passage, before the sentence — the relevance judgment conditions what is written; the model learns to lean less on irrelevant passages

IsSup after the sentence — a post-hoc entailment check of yt against d; it cannot come earlier

IsUse at the very end — utility is a property of the whole answer

What the model writes (paper Fig. 2, Table 4)

Q: when did Walking Dead season 7 come out? (Natural Questions)

[Retrieve=Yes] ...season 7 premiered on October 23, 2016... [IsRel=Relevant] October 23, 2016 [IsSup=Fully] [IsUse=5]

Q: Synthesize a poem: 'The sun is setting' (Stanford Alpaca)

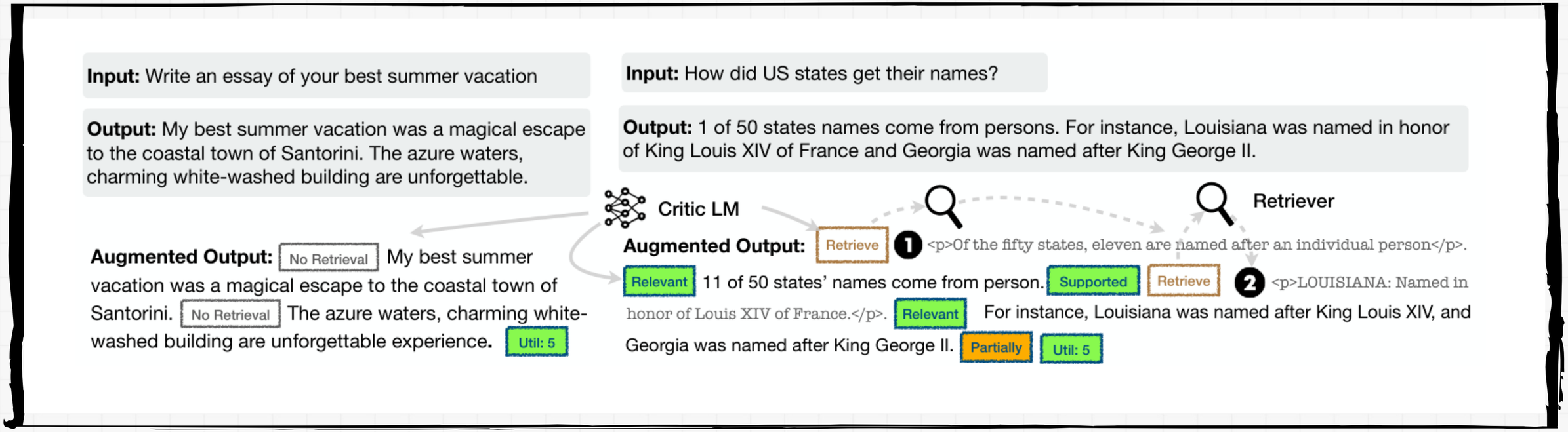
[Retrieve=No] The stars begin to twinkle bright... [Retrieve=No] They flicker like a silent dance... [IsUse=5]

Q: Give an example of a situation where you delegate tasks (GPT-4 Alpaca)

[Retrieve=No] The manager realises... [Retrieve=Yes] Delegation... [IsRel=Relevant] This includes creating a guest list... [IsSup=Partially] ... [Retrieve=Yes] Joint application design... [IsRel=Irrelevant] ... [IsUse=5]

From the paper: Figure 2 (training examples)

Asai et al. 2024, Fig. 2



- Left: no retrieval needed; only [No Retrieval] tokens and a final [Util: 5].
- Right: the critic LM inserts [Retrieve], passages, [Relevant], [Supported] / [Partially] around the original output; the output text itself is unchanged.

Vocabulary expansion and the restricted softmax

Adding reflection tokens R

$$E \in \mathbb{R}^{|V| \times h} \rightarrow E' \in \mathbb{R}^{(|V|+|R|) \times h}, \qquad W \in \mathbb{R}^{|V| \times h} \rightarrow W' \in \mathbb{R}^{(|V|+|R|) \times h}$$

E: input embedding table; W: output head. New rows are learned in fine-tuning; a reflection token is then emitted exactly like a word, through the same softmax.

Restricted softmax: the identity behind every score

$$p(r \mid \text{next} \in G) = \frac{p(r)}{\sum_{r' \in G} p(r')} = \frac{e^{z_r}/Z}{\sum_{r' \in G} e^{z_{r'}}/Z} = \frac{e^{z_r}}{\sum_{r' \in G} e^{z_{r'}}}$$

Z (sum over all ~32k tokens) cancels. A group's probabilities can be read off the logits of that group alone, ignoring mass on unrelated words.

Tokens in the released models

[Retrieval] · [No Retrieval] · [Continue to Use Evidence]

[Relevant] · [Irrelevant]

[Fully supported] · [Partially supported] · [No support / Contradictory]

[Utility:1] ... [Utility:5]

<paragraph> ... </paragraph>

Training overview (Algorithm 2)

GPT-4 labels → Critic C → Annotate 150k → Generator M

Few-shot prompts per token type: 12.6k Retrieve, 19.3k IsRel, 11.2k IsSup, 3.8k IsUse

Llama2-7B fine-tuned on those labels (Eq. 1). Agrees with GPT-4 74-94% depending on type

For each sentence: C decides retrieval, R fetches, C labels IsRel / IsSup, keeps one passage; IsUse at the end

Llama2 7B / 13B with new tokens; next-token loss on the annotated corpus (Eq. 2)

At inference: only M + the retriever

C and GPT-4 are only used to build training data. Two distillation steps: GPT-4 → C → M. M ends up being its own critic.



Stage 1a: critic labels from GPT-4

- **Why not label everything with GPT-4?** cost (sentences × passages × 4 aspects per example) and reproducibility (API drift).
- **Sampling:** for each token type, sample (x, y) pairs from the training set.
- **Prompt:** type-specific instruction + few-shot demonstrations; GPT-4 outputs the label as text, sampling r from p(r | I, x, y).
- **Decoding:** temperature 1, ≤ 200 tokens, '##' separator; outputs in the wrong format are discarded.

	Retrieve	IsRel	IsSup	IsUse
Labels kept	12,594	19,317	11,181	3,831
Author agreement (n = 20)	95%	95%	90%	80%

Example prompt: Retrieve, input only (Table 8)

'Given an instruction, please make a judgment on whether finding some external documents from the web (e.g., Wikipedia) helps to generate a better response. Please answer [Yes] or [No] and write an explanation.'

Give three tips for staying healthy. **[Yes]** reliable sources can support the tips

Describe a time you made a difficult decision. **[No]** personal experience

Find the area of a circle, radius = 4. **[No]** maths; no document states the answer

Explain cellular respiration in plants. **[Yes]** detailed scientific concept

Stage 1b: training the critic (Eq. 1)

$$\max_{C} \; \mathbb{E}_{((x,y),r)\sim\mathcal{D}_{critic}} \log p_C(r \mid x, y)$$

- **Meaning:** make the critic put high probability on the label GPT-4 gave.
- **Equivalently:** cross-entropy between C's distribution over label tokens and GPT-4's one-hot label; a classifier implemented as conditional LM.
- **Initialisation:** Llama2-7B, the same base as the generator.
- **Inputs:** (x, y) plus the passage d for IsRel / IsSup.

Agreement with GPT-4 on held-out labels (App. Table 5)

Base LM	Retrieve	IsSup	IsRel	IsUse
Llama2-7B	93.8	93.5	80.2	73.5
FLAN-3B	85.6	73.1	82.0	72.1

Bigger base model helps on most types. Both struggle most on IsUse, mainly confusing 4 and 5, where human annotators also disagree. These are agreement rates with GPT-4, not with humans.

Stage 2a: annotating the corpus (Alg. 3)

STEP 1	Retrieve?	C predicts Retrieve for (x, y). If No: only predict IsUse; done.
STEP 2	Initial retrieval	If Yes: retrieve with (x, whole y) as query; split y into sentences.
STEP 3	Per sentence	C decides Yes / No / Continue given x, y<t and the earlier passage; if Yes, query R with (x, yt).
STEP 4	Judge passages	For each retrieved d: C predicts IsRel, then IsSup.
STEP 5	Pick one passage	Rule on the right; insert it with its tokens.
STEP 6	Utility	At the end C predicts IsUse; add the sequence to D_gen.

Passage selection rule

- Keep d with IsRel = Relevant and IsSup = Fully or Partially
- Several qualify: highest retrieval score
- None qualify: pick one at random

Why random? It keeps negative examples, so M learns to emit Irrelevant / No support honestly.

The answer text y is never rewritten; only tokens and passages are inserted.

Training data: 145,619 examples (App. Table 3)

Dataset	Category	Source	Instances
GPT-4 Alpaca	Instruction-following	Open-Instruct	26,168
Stanford Alpaca	Instruction-following	Open-Instruct	25,153
FLAN-V2	Instruction-following	Open-Instruct	17,817
ShareGPT	Instruction-following	Open-Instruct	13,406
Open Assistant 1	Instruction-following	Open-Instruct	9,464
Wizard of Wikipedia	Knowledge-intensive	KILT	17,367
Natural Questions	Knowledge-intensive	KILT	15,535
FEVER	Knowledge-intensive	KILT	9,966
OpenBookQA	Knowledge-intensive	HF Datasets	4,699
ARC-Easy	Knowledge-intensive	HF Datasets	2,147
ASQA	Knowledge-intensive	ASQA	3,897

Mix

92,008

instruction-following (63%)

53,611

knowledge-intensive (37%)

Diverse instructions keep versatility; knowledge-intensive sets teach when retrieval pays off. Ablations later use a 50k random subset.

Stage 2b: training the generator (Eq. 2)

$$\max_{\mathcal{M}} \; \mathbb{E}_{(x,y,r)\sim\mathcal{D}_{gen}} \log p_{\mathcal{M}}(y, r \mid x)$$

$$\mathcal{L}_{gen} = -\sum_{i=1}^{n} m_i \log p_{\mathcal{M}}(\tilde{w}_i \mid x, \tilde{w}_{<i}), \qquad m_i = \mathbf{1}\left[\tilde{w}_i \notin \langle p\rangle \ldots \langle /p\rangle\right]$$

Versus Eq. 1

- Critic predicts only r given (x, y)
- Generator predicts y AND r, interleaved
- Vocabulary expanded with all reflection tokens

Why mask passages?

- They are inputs from R, not model output
- Predicting them = memorising Wikipedia, or inventing passage-like text
- Still attended to: the model learns to use them

Effect

- M learns the retrieval policy, the critique and the writing in one model
- At inference M's own token probabilities act as reward estimates

Why not RLHF? Cost, stability, control

	PPO-based RLHF	Self-RAG
Models in GPU memory	policy, reference, reward, value (about 4)	1 (generator)
Data	on-policy samples during training	static corpus, labelled once
Objective	policy gradient + KL penalty	cross-entropy
Reward model at inference	no (baked in)	no (M emits it)
Change behaviour later	retrain	change w, δ at inference

Trade-off: RLHF optimises the policy itself; Self-RAG only learns to predict critique and leaves selection to decoding. Cheaper and controllable, but the generator never directly improves from the reward.

Training setup (App. B.1)

- 4 × A100 80GB; 3 epochs
- Batch 128; peak LR 2e-5, 3% warm-up, linear decay
- Max length 2,048 (7B) / 1,524 (13B)
- DeepSpeed ZeRO-3, bf16, FlashAttention
- Inference on 1-2 Quadro RTX 6000 (24GB) with vLLM

Inference: Algorithm 1

Require: generator M, retriever R, passages {d1, ..., dN}

Input: prompt x, previous segments y<t; output: segment yt

```
1  M predicts Retrieve given (x, y<t)
2  if Retrieve = Yes (or p_yes > δ):
3      D ← R(x, yt−1)    retrieve K passages
4      for each d in D, in parallel:
5          M predicts IsRel given (x, d)
6          M predicts yt given (x, d, y<t)
7          M predicts IsSup, IsUse given (x, yt, d)
8      rank the K candidates by IsRel, IsSup, IsUse
9  else:
10     M predicts yt given (x, y<t), then IsUse
```

Defaults (§4.3)

- K = 5 passages per retrieval
- Beam width B = 2 over segments
- Greedy decoding inside a segment
- w = (1.0, 1.0, 0.5) for IsRel, IsSup, IsUse
- δ = 0.2 (δ = 0 for ASQA)
- vLLM for parallel branches

Adaptive retrieval threshold (App. A.3)

$$\text{retrieve at step } t \iff \frac{p(\text{Yes})}{p(\text{Yes}) + p(\text{No})} > \delta$$

- **Restricted softmax:** probability of Yes given the model is choosing between Yes and No.
- **δ → 0:** retrieve almost always, like standard RAG. Used for ASQA, where citations are required.
- **δ → 1:** almost never retrieve.
- **δ = 0.2 (default):** retrieve even when No is more likely, as long as Yes keeps ≥ 20% of the Yes/No mass. Biased towards retrieving on factual tasks.
- **Hard constraint:** retrieve only if Yes is the argmax, i.e. effectively δ = 0.5.

Worked example

p(Yes) = 0.12, p(No) = 0.40, p(Continue) = 0.03, the rest on other tokens

$$\frac{0.12}{0.12 + 0.40} = 0.231 > 0.2 \Rightarrow \text{retrieve}$$

Hard constraint: argmax is No (0.40 > 0.12), so no retrieval. This is why 'hard constraints' retrieves less and loses 17 points on PopQA in the ablation (45.5 → 28.3).

Turning critique tokens into scores

$$s^{Rel} = \frac{p(\text{Relevant})}{p(\text{Relevant}) + p(\text{Irrelevant})} \in [0, 1]$$

$$s^{Sup} = \frac{p(\text{Fully})}{S} + 0.5 \cdot \frac{p(\text{Partially})}{S}, \quad S = p(\text{Fully}) + p(\text{Partially}) + p(\text{No})$$

$$s^{Use} = \sum_{i=1}^{5} u_i \frac{p(\text{IsUse} = i)}{S}, \quad u = (-1, -0.5, 0, 0.5, 1), \quad S = \sum_{j=1}^{5} p(\text{IsUse} = j) \in [-1, 1]$$

Unified view: expected value

$$s^{G} = \mathbb{E}_{r \sim p(\cdot \mid r \in G)}\left[v_G(r)\right] = \sum_{r \in G} v_G(r) \frac{p(r)}{\sum_{r' \in G} p(r')}$$

v_Rel = (1, 0); v_Sup = (1, 0.5, 0); v_Use = (−1, −0.5, 0, 0.5, 1). The paper's generic formula p(r̂) / Σ p(r) is the special case where only the best value counts.

Why normalise?

- Raw p(Relevant) also competes with ~32k ordinary tokens
- Renormalising removes that nuisance mass
- Fixed ranges allow a weighted sum

Segment score and beam search (Eqs. 3-4)

$$f(y_t, d, \text{Critique}) = p(y_t \mid x, d, y_{<t}) + \mathcal{S}(\text{Critique})$$

$$\mathcal{S}(\text{Critique}) = \sum_{G \in \mathcal{G}} w^G s_t^G, \qquad \mathcal{G} = \{\text{IsRel}, \text{IsSup}, \text{IsUse}\}$$

- **First term:** how likely the segment is under the LM; in the released code a length-normalised probability exp(mean log p), so it lies in [0, 1].
- **Second term:** weighted linear reward from self-critique. Defaults w = (1, 1, 0.5).
- **Hard alternative:** drop any candidate whose IsSup is No support.
- **Closed-set tasks:** scores are aggregated per answer option; the best option wins (App. B.2).

Segment-level beam search

- Keep B = 2 partial answers
- Expand each into K candidates when retrieving (one per passage), else one
- Score candidates with f, accumulated along the path
- Keep the top B; repeat until all beams finish; return the best
- Tokens inside a segment are decoded greedily

Worked example: why the less fluent candidate wins

Query: 'How did US states get their names?'	d1: 11 of 50 states named after people	d2: popular baby names in Texas
Fluency p(yt | ·)	0.60	0.70
p(Rel), p(Irr) → s Rel	0.9, 0.1 → 0.90	0.2, 0.8 → 0.20
p(Full), p(Part), p(No) → s Sup	0.7, 0.2, 0.1 → 0.80	0.1, 0.3, 0.6 → 0.25
p(Use = 1…5) → s Use	.01 .04 .15 .40 .40 → 0.57	.05 .10 .25 .35 .25 → 0.325
f = p + s Rel + s Sup + 0.5 s Use	2.585 (chosen)	1.3125

Illustrative numbers, not from the paper.

Sensitivity to the weights

$$f(d_1) - f(d_2) = 0.5725 + 0.70\, w^{Rel}$$

With w Sup = 1 and w Use = 0.5, d2 wins only if w Rel < −0.818, i.e. only if irrelevance is rewarded. Set all w = 0 and ranking reduces to likelihood, which picks d2.

Takeaway: critique terms move selection from 'most fluent' to 'most fluent and grounded'.

Inference cost

$$N_{gen} \approx \sum_{t=1}^{T} B[K \cdot \mathbf{1}[ret_t] + (1 - \mathbf{1}[ret_t])]$$

Number of segment generations. With B = 2 and K = 5, a retrieval step costs up to 10 drafts, each with its own passage prefill (~100 words), plus a few critique tokens.

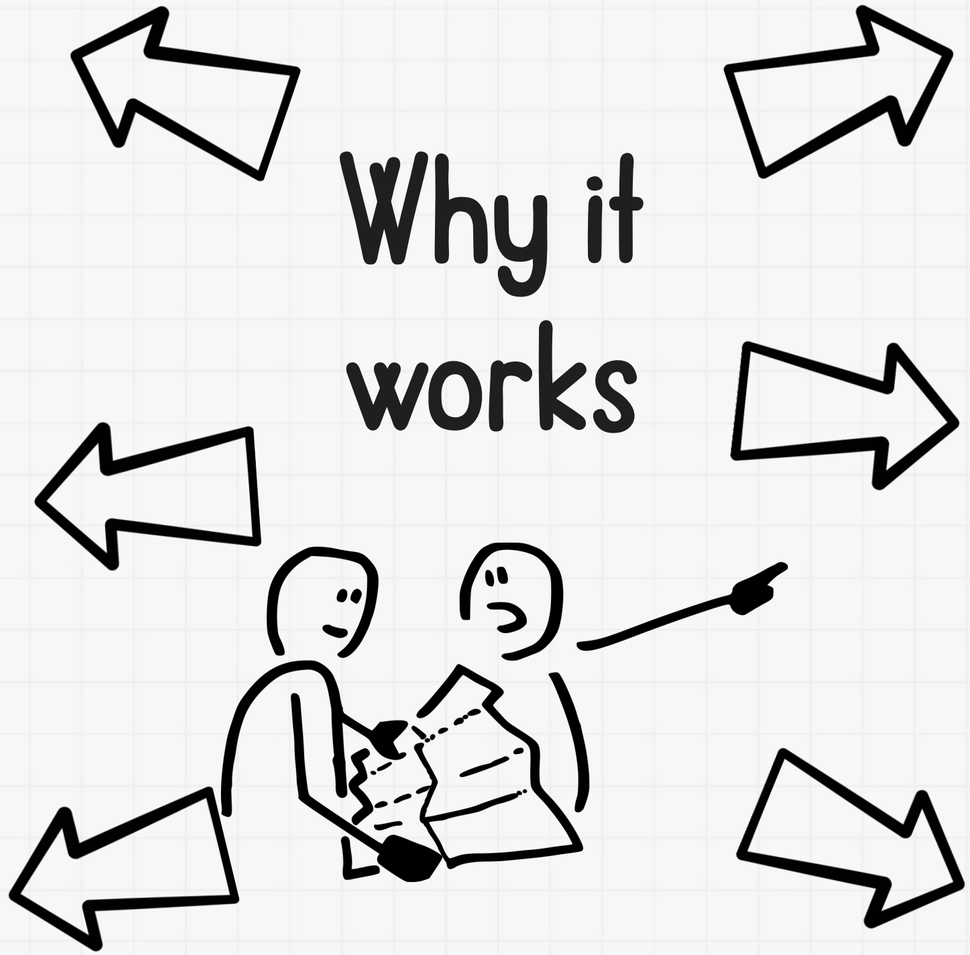
What makes it expensive

- B · K branches per retrieval step
- Prefill of a passage in each branch
- Retrieval calls (dense search) per step
- Latency and throughput are not reported in the paper

What keeps it manageable

- Branches are independent, so they batch well (vLLM)
- Greedy decoding inside segments
- No retrieval → a single branch (creative prompts)
- δ trades accuracy for retrieval frequency (Fig. 3c)

Why it works



Decomposition

4 small, separately supervised decisions instead of one hard one

Retrieve if useful

Essays and maths stay a normal LM, so versatility is kept

Generate + verify

Checking a sentence against one passage is easier than writing a correct one

Cheap critique

Learned offline with SFT; no RL, no reward model at test time

One passage each

No mixing of noisy passages; bad ones are scored down

RLHF in disguise

Ranking by likelihood + $\Sigma\, w \cdot s$ mimics a reward-tilted policy (next slide)

A closer look: test-time reward tilting

KL-regularised RLHF objective

$$\max_{\pi} \; \mathbb{E}_{y \sim \pi(\cdot|x)}[r(x,y)] - \beta \, \mathrm{KL}(\pi(\cdot|x) \,\|\, \pi_{ref}(\cdot|x))$$

Closed-form optimum

$$\pi^*(y|x) = \frac{1}{Z(x)} \, \pi_{ref}(y|x) \exp\left(\frac{r(x,y)}{\beta}\right)$$

$$\log \pi^*(y|x) = \log \pi_{ref}(y|x) + \tfrac{1}{\beta} r(x,y) - \log Z(x)$$

Self-RAG ranking

$$\mathrm{score}(y_t) = p(y_t|\,\cdot\,) + \sum_G w^G s_t^G$$

- Same shape as log π* with reward r = Σ w · s and β = 1, over a finite candidate set
- Changing w = choosing a different reward, with no retraining
- **Caveat:** Self-RAG adds p, not log p, so this is an analogy, not an equivalence

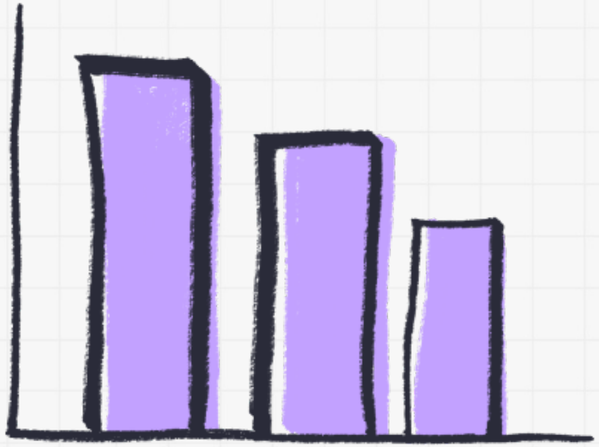
Contrast with Lewis-RAG

$$\text{Lewis}: \sum_z p_\eta(z|x) \, p_\theta(y|x,z) \qquad \text{Self-RAG}: \; \arg\max_{d \in D} \left[p(y_t|x,d) + \sum_G w^G s^G(x,d,y_t) \right]$$

Soft average weighted by the retriever vs hard selection weighted by the generator's own judgment.

04

Experiments



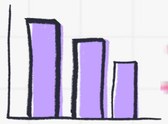
- Six tasks, metrics with formulas
- Baselines and fairness details
- Full Table 2 and takeaways
- Ablations, test-time control, efficiency, data scale
- Human evaluation and qualitative examples

Tasks and datasets (§4.1), all zero-shot



Closed-set

- **PubHealth:** public-health fact verification (Zhang 2023)
- **ARC-Challenge:** science exam multiple choice (Clark 2018)
- **Metric:** accuracy; answer-option probabilities aggregated



Short-form QA

- **PopQA long-tail:** 1,399 rare-entity questions (<100 monthly views)
- **TriviaQA-unfiltered:** 11,313 test questions (Min 2019 split)
- **Metric:** is a gold answer contained in the output



Long-form

- **Biographies:** 'Tell me a bio about X', FactScore (Min 2023)
- **ASQA (ALCE):** ambiguous questions; long answers with citations
- **Metrics:** str-em, ROUGE, MAUVE, citation precision and recall

Metrics (1/2): accuracy, match, FactScore

Match accuracy (PopQA, TQA)

$$\mathrm{Acc} = \frac{1}{N}\sum_{j=1}^{N} \mathbf{1}\left[\exists\, a \in \mathcal{A}_j :\ a \subseteq \hat{y}_j\right]$$

A_j: gold aliases. Lenient by design (Mallen 2023; Schick 2023). Caveat: a long answer listing many candidates can be 'correct' by accident.

FactScore (Bio)

$$\mathrm{FS}(y) = \frac{1}{|\mathcal{A}_y|}\sum_{a \in \mathcal{A}_y} \mathbf{1}[a \text{ supported by Wikipedia}]$$

$$\mathrm{FS} = \mathbb{E}_x[\mathrm{FS}(y_x)]$$

A_y: atomic facts extracted by an LLM, each checked by retrieval + LM verification. Measures precision, not coverage.

- **Closed-set accuracy:** for PubHealth and ARC, Self-RAG aggregates scores per option and picks the best option; predictions are lower-cased and string-matched to avoid format errors (App. B.2).
- **Why it matters:** a 3-fact bio with 3 supported facts scores 1.0; a 10-fact bio with 8 supported scores 0.8. Shorter answers can win on FactScore.

Metrics (2/2): ASQA correctness, fluency, citations

Correctness

- str-em: fraction of the disambiguated short answers that appear verbatim in the output
- ROUGE-L: longest-common-subsequence overlap with reference long answers

Fluency: MAUVE

$$R_\lambda = \lambda P + (1-\lambda)Q$$

$$\mathcal{C}(P,Q) = \left\{ \left(e^{-c\,\mathrm{KL}(Q\|R_\lambda)},\ e^{-c\,\mathrm{KL}(P\|R_\lambda)} \right) : \lambda \in (0,1) \right\}$$

MAUVE = area under C (P human, Q model text, quantised embeddings); 1 = identical.

Citations (ALCE; NLI judge TRUE, T5-11B)

$$\mathrm{rec}_i = \mathbf{1}[\mathrm{concat}(C_i) \models s_i]$$

$$c \in C_i \text{ irrelevant} \Leftrightarrow (\not\models s_i) \wedge \left(C_i \setminus \{c\} \models s_i \right)$$

Precision = share of citations that are not irrelevant. Recall: are claims backed? Precision: does each citation pull its weight?

Fairness fixes in the paper

- All systems capped at 100 new tokens on ASQA (Llama2-chat's long outputs inflated str-em)
- Same author-provided GTR-XXL top-5 passages for every system

Baselines (§4.2) and what each one controls for

Group	Models	Controls for
No retrieval, public	Llama2 7B/13B; Alpaca 7B/13B (Llama2 replication)	the effect of retrieval + reflection
No retrieval, RLHF / private data	ChatGPT; Llama2-chat-13B	is a 7B Self-RAG competitive with larger RLHF models?
Prompt engineering	CoVE-65B (chain-of-verification)	iterative self-verification by prompting vs learned reflection
Standard RAG	Llama2, Alpaca, Llama2-chat, ChatGPT + the same retrieved passages	the key comparison: same retriever, no reflection
Same data, no reflection	Llama2-FT-7B: trained on the 150k pairs without tokens or passages	is the gain from data or from the framework?
Trained with retrieval	SAIL-7B; Toolformer-6B (numbers from their papers)	other retrieval-aware training
Production system	Perplexity.ai (Bio only)	a deployed search-LLM product

Evaluation settings and fairness

Retrieval sources

- Contriever-MS MARCO, top-5 from Wikipedia (official 2018 embeddings)
- PopQA: Dec 2020 Wikipedia (Izacard 2022b), since 2018 lacks newer entities
- Bio and open-domain QA: +5 documents from Google Programmable Search (intro paragraphs), following SAIL
- ASQA: author-provided GTR-XXL top-5 for all baselines

Generation settings

- Open QA: ≤ 100 new tokens; closed-set: ≤ 50
- FactScore: ≤ 500 for baselines, ≤ 200 per segment for Self-RAG
- ALCE capped at 100 tokens for all systems (baselines then output 30–60 tokens)
- Official chat templates for instruction-tuned baselines
- Fact-check labels lower-cased; MCQ formats string-matched

Table 2 (1/2): vs proprietary and no-retrieval models

Model	PopQA	TQA	Pub	ARC	Bio FS	ASQA em	rg	mau	prec	rec
Llama2-chat-13B	20.0	59.3	49.4	38.4	55.9	22.4	29.6	28.6	-	-
Ret-Llama2-chat-13B	51.8	59.8	52.1	37.9	79.9	32.8	34.8	43.8	19.8	36.1
ChatGPT	29.3	74.3	70.1	75.3	71.8	35.3	36.2	68.8	-	-
Ret-ChatGPT	50.8	65.7	54.7	75.3	-	40.7	39.9	79.7	65.1	76.6
Perplexity.ai	-	-	-	-	71.2	-	-	-	-	-
Llama2-7B	14.7	30.5	34.2	21.8	44.5	7.9	15.3	19.0	-	-
Alpaca-7B	23.6	54.5	49.8	45.0	45.8	18.8	29.4	61.7	-	-
Llama2-13B	14.7	38.5	29.4	29.4	53.4	7.2	12.4	16.0	-	-
Alpaca-13B	24.4	61.3	55.5	54.9	50.2	22.9	32.0	70.6	-	-
CoVE-65B *	-	-	-	-	71.2	-	-	-	-	-
Self-RAG 7B	54.9	66.4	72.4	67.3	**81.2**	30.0	35.7	74.3	66.9	67.8
Self-RAG 13B	**55.8**	69.3	**74.5**	73.1	80.2	31.7	37.0	71.6	**70.3**	71.3

Yellow: Self-RAG. Pink: a proprietary model beats Self-RAG 13B. * concurrent work. rg ROUGE, mau MAUVE, prec/rec citations.

Table 2 (2/2): vs retrieval-augmented baselines

Model	PopQA	TQA	Pub	ARC	Bio FS	ASQA em	rg	mau	prec	rec
Toolformer-6B *	-	48.8	-	-	-	-	-	-	-	-
Llama2-7B + RAG	38.2	42.5	30.0	48.0	78.0	15.2	22.1	32.0	2.9	4.0
Alpaca-7B + RAG	46.7	64.1	40.2	48.0	76.6	30.9	33.3	57.9	5.5	7.2
Llama2-FT-7B + RAG	48.7	57.3	64.3	65.8	78.2	31.0	35.8	51.2	5.0	7.5
SAIL-7B *	-	-	69.2	48.4	-	-	-	-	-	-
Llama2-13B + RAG	45.7	47.0	30.2	26.0	77.5	16.3	20.5	24.7	2.3	3.6
Alpaca-13B + RAG	46.1	66.9	51.1	57.6	77.7	**34.8**	36.7	56.6	2.0	3.8
Self-RAG 7B	54.9	66.4	72.4	67.3	**81.2**	30.0	35.7	**74.3**	66.9	67.8
Self-RAG 13B	**55.8**	**69.3**	**74.5**	**73.1**	80.2	31.7	**37.0**	71.6	**70.3**	**71.3**

Bold: best among non-proprietary models. Teal: Llama2-FT, the same-data control. * numbers reported in the original papers (no implementation). Alpaca-13B + RAG has the best open str-em (34.8); Self-RAG leads every other column.

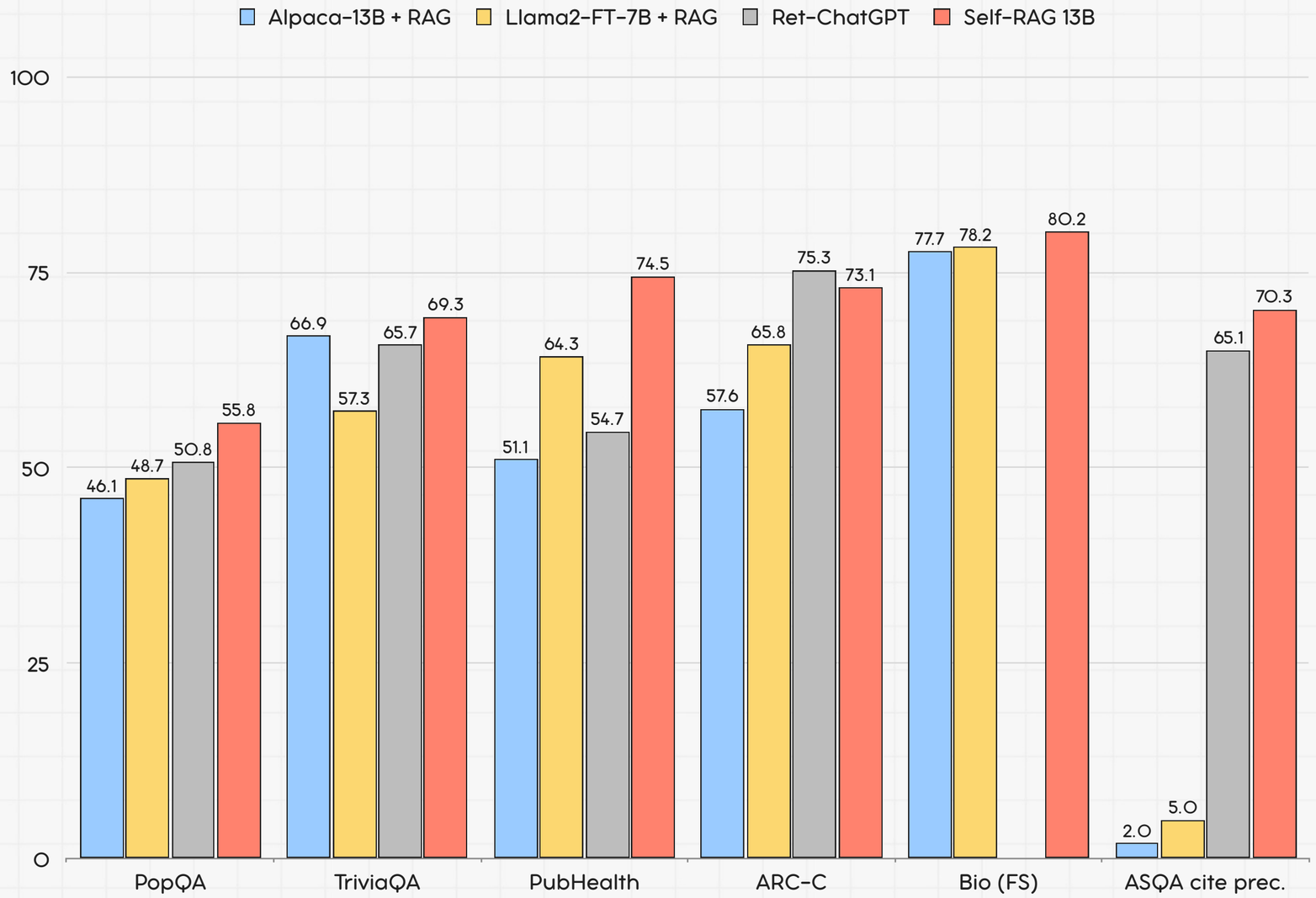
From the paper: Table 2

Asai et al. 2024, Table 2

LM	Short-form PopQA (acc)	Short-form TQA (acc)	Closed-set Pub (acc)	Closed-set ARC (acc)	Long-form generations (with citations) Bio (FS)	ASQA (em)	ASQA (rg)	ASQA (mau)	ASQA (pre)	ASQA (rec)
LMs with proprietary data										
Llama2-c$_{13B}$	20.0	59.3	49.4	38.4	55.9	22.4	29.6	28.6	–	–
Ret-Llama2-c$_{13B}$	51.8	59.8	52.1	37.9	79.9	32.8	34.8	43.8	19.8	36.1
ChatGPT	29.3	**74.3**	70.1	**75.3**	71.8	35.3	36.2	68.8	–	–
Ret-ChatGPT	50.8	65.7	54.7	**75.3**	–	**40.7**	**39.9**	**79.7**	65.1	**76.6**
Perplexity.ai	–	–	–	–	71.2	–	–	–	–	–
Baselines without retrieval										
Llama2$_{7B}$	14.7	30.5	34.2	21.8	44.5	7.9	15.3	19.0	–	–
Alpaca$_{7B}$	23.6	54.5	49.8	45.0	45.8	18.8	29.4	61.7	–	–
Llama2$_{13B}$	14.7	38.5	29.4	29.4	53.4	7.2	12.4	16.0	–	–
Alpaca$_{13B}$	24.4	61.3	55.5	54.9	50.2	22.9	32.0	70.6	–	–
CoVE$_{65B}$ *	–	–	–	–	71.2	–	–	–	–	–
Baselines with retrieval										
Toolformer*$_{6B}$	–	48.8	–	–	–	–	–	–	–	–
Llama2$_{7B}$	38.2	42.5	30.0	48.0	78.0	15.2	22.1	32.0	2.9	4.0
Alpaca$_{7B}$	46.7	64.1	40.2	48.0	76.6	30.9	33.3	57.9	5.5	7.2
Llama2-FT$_{7B}$	48.7	57.3	64.3	65.8	78.2	31.0	35.8	51.2	5.0	7.5
SAIL*$_{7B}$	–	–	69.2	48.4	–	–	–	–	–	–
Llama2$_{13B}$	45.7	47.0	30.2	26.0	77.5	16.3	20.5	24.7	2.3	3.6
Alpaca$_{13B}$	46.1	66.9	51.1	57.6	77.7	**34.8**	36.7	56.6	2.0	3.8
Our SELF-RAG $_{7B}$	54.9	66.4	72.4	67.3	**81.2**	30.0	35.7	**74.3**	66.9	67.8
Our SELF-RAG $_{13B}$	**55.8**	**69.3**	**74.5**	**73.1**	80.2	31.7	**37.0**	71.6	**70.3**	**71.3**

- **Bold**: best non-proprietary model; grey bold: proprietary model better than all open ones. * numbers from concurrent work.

Main results at a glance



- **Best open model** on all six tasks
- **Beats ChatGPT** on PopQA, PubHealth, Bio and citation precision
- **Not everywhere:** ChatGPT without retrieval still wins TriviaQA (74.3) and ARC (75.3)
- **Same data is not enough:** Llama2-FT trails, so the gain comes from the framework

Source: Table 2. Ret-ChatGPT has no Bio score.

Reading Table 2: five takeaways

1 Standard RAG can hurt

Ret-ChatGPT PubHealth 70.1 → 54.7; Alpaca-13B 55.5 → 51.1; ARC barely moves. Retrieval helps where answers can be copied (PopQA, Bio) but not for verification or reasoning.

2 Citations jump >10×

Open RAG models reach 2–8% citation precision/recall; Self-RAG 13B reaches 70.3 / 71.3, above Ret-ChatGPT on precision (65.1), below it on recall (76.6).

3 Framework, not data

Llama2-FT-7B (same 150k data): PubHealth 64.3 vs 72.4, citation precision 5.0 vs 66.9.

4 Small beats large

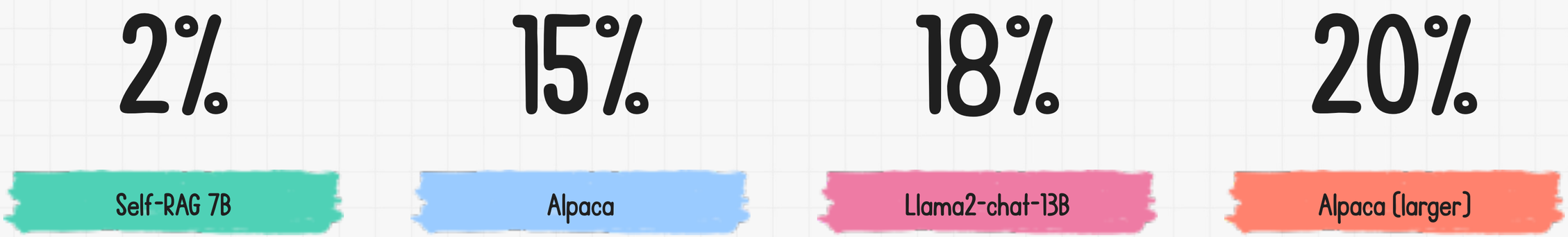
Self-RAG 7B/13B beats CoVE-65B on Bio (81.2 vs 71.2) and Llama2-chat-13B everywhere.

5 7B vs 13B

7B sometimes wins on factual precision (Bio 81.2 vs 80.2): it writes shorter, precisely grounded outputs.

Do answers come from the evidence? (App. C.1)

Among correct answers on PopQA and TriviaQA: how often is the gold answer NOT a substring of the retrieved passages, i.e. came from parametric memory?



When passages are irrelevant, Self-RAG emits IsRel = Irrelevant, signalling that the following answer may not be grounded, while instruction-tuned baselines keep producing plausible answers. The paper's text names the largest baseline 'Alpaca 30B', which does not appear elsewhere; treat that label with care.

Ablations (Fig. 3a): every component matters

Variant (7B, 50k training examples)	PopQA	PubHealth	ASQA em	What it shows
Self-RAG (full)	45.5	73.5	32.1	reference
Train: no retriever (plain instruction tuning)	43.6	67.8	31.0	training with passages helps, most on PubHealth
Train: no critic (always top-1, no tokens; like SAIL)	42.6	72.0	18.1	reflection tokens are the core of long-form gains
Test: never retrieve	24.7	73.0	-	long-tail facts need retrieval; PubHealth does not
Test: hard constraint (argmax Retrieve)	28.3	72.6	-	retrieves too rarely: δ is a crucial knob
Test: always top-1 passage (standard RAG)	41.8	73.1	28.6	choosing among K by critique beats trusting R
Test: drop IsSup from the score	44.1	73.2	30.6	support scoring matters most for long-form

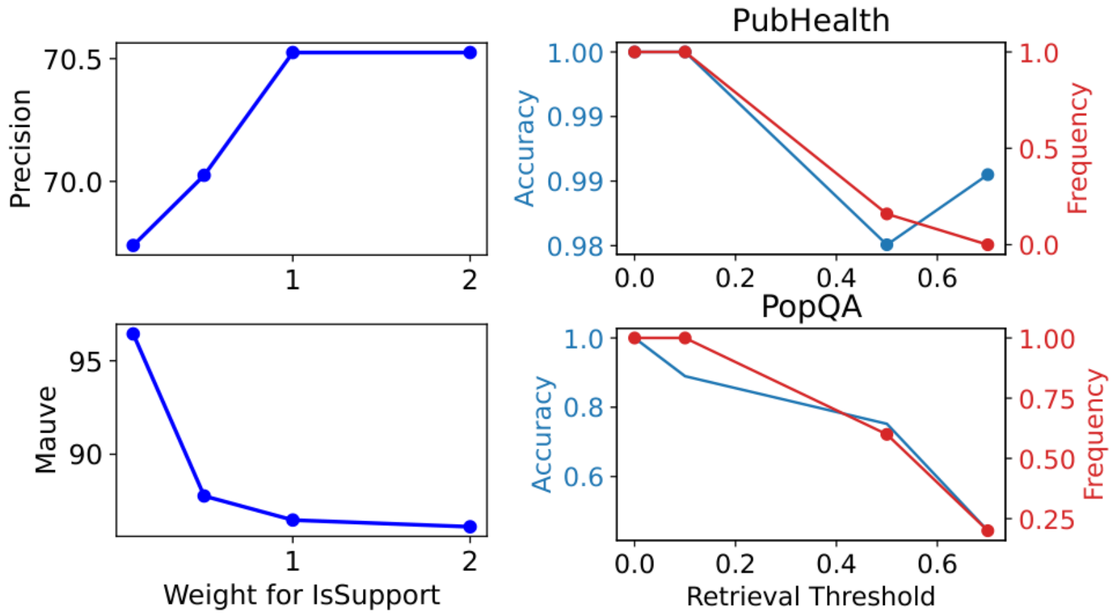
ASQA ablations use 150 sampled instances and exclude adaptive / no-retrieval variants. No variance or seeds are reported.

From the paper: Figure 3

Asai et al. 2024, Fig. 3

	PQA (acc)	Med (acc)	AS (em)
SELF-RAG (50k)	45.5	73.5	32.1
Training			
No Retriever $\mathcal{R}$	43.6	67.8	31.0
No Critic $\mathcal{C}$	42.6	72.0	18.1
Test			
No retrieval	24.7	73.0	–
Hard constraints	28.3	72.6	–
Retrieve top1	41.8	73.1	28.6
Remove IsSup	44.1	73.2	30.6

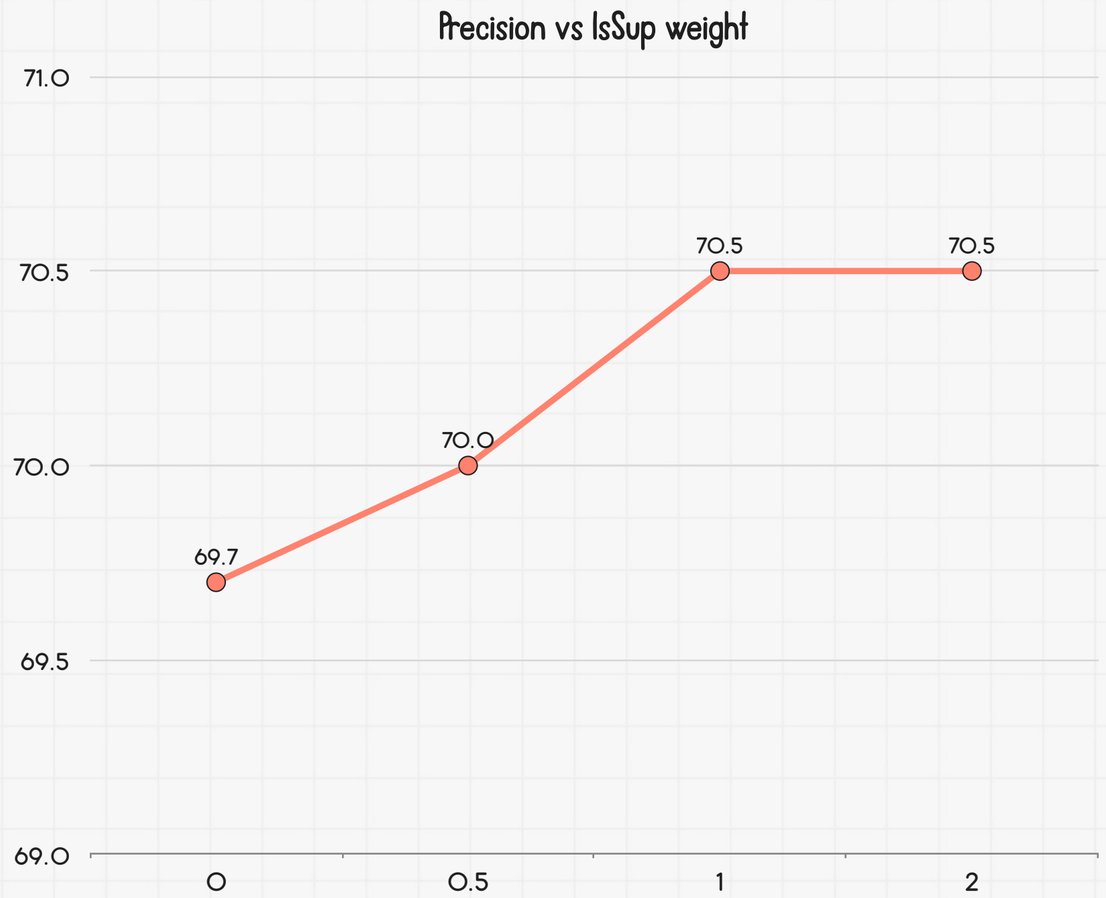
(a) Ablation

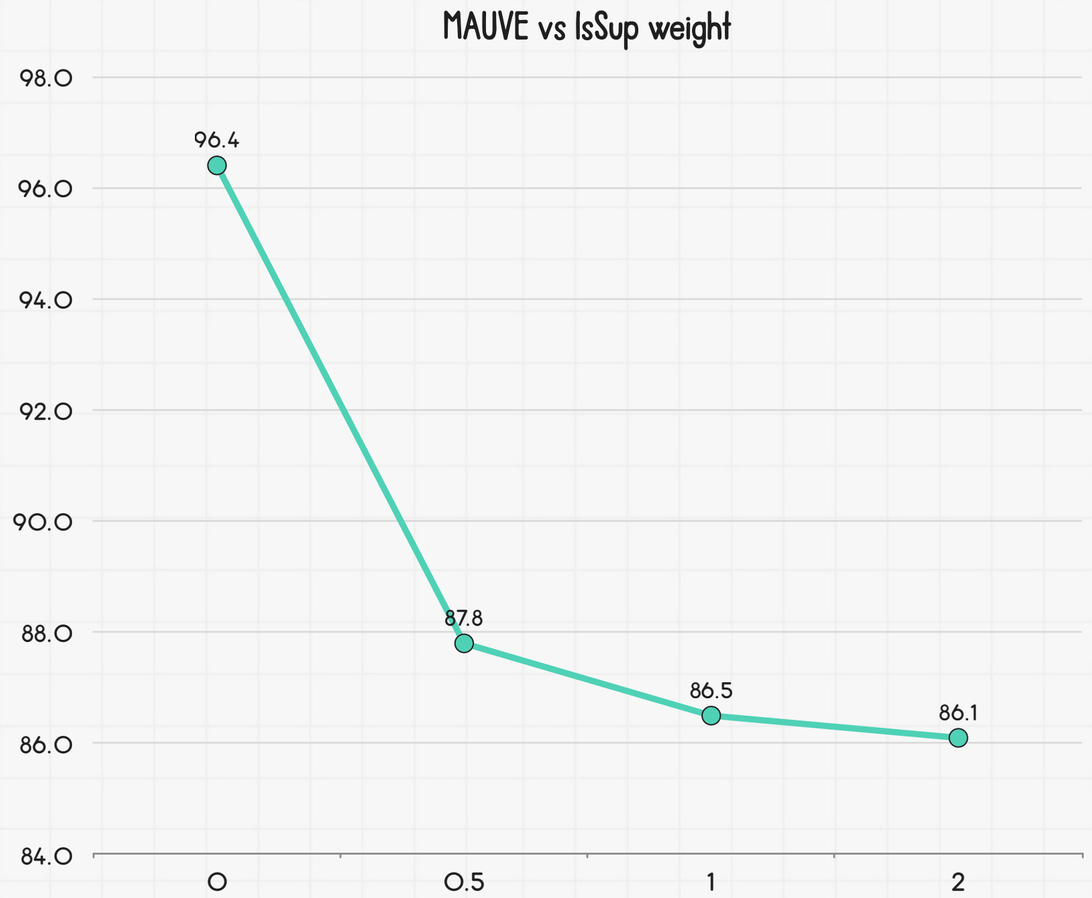


(b) Customization (c) Retrieval

- (a) Ablation: PQA = PopQA, Med = PubHealth, AS = ASQA (em); 7B model trained on 50k examples.
- (b) Customisation: IsSup weight vs citation precision (top) and MAUVE (bottom) on ASQA.
- (c) Retrieval: threshold δ vs normalised accuracy (blue) and retrieval frequency (red) on PubHealth and PopQA.

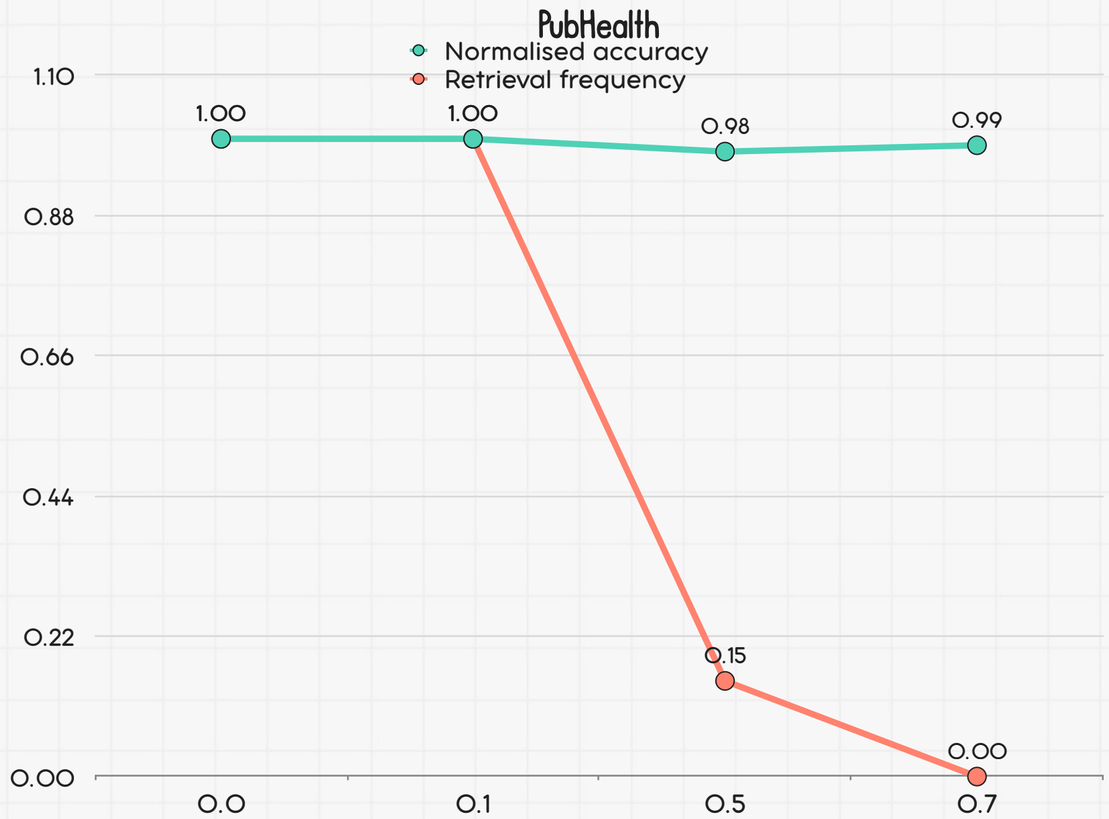
Test-time customisation (Fig. 3b, ASQA)

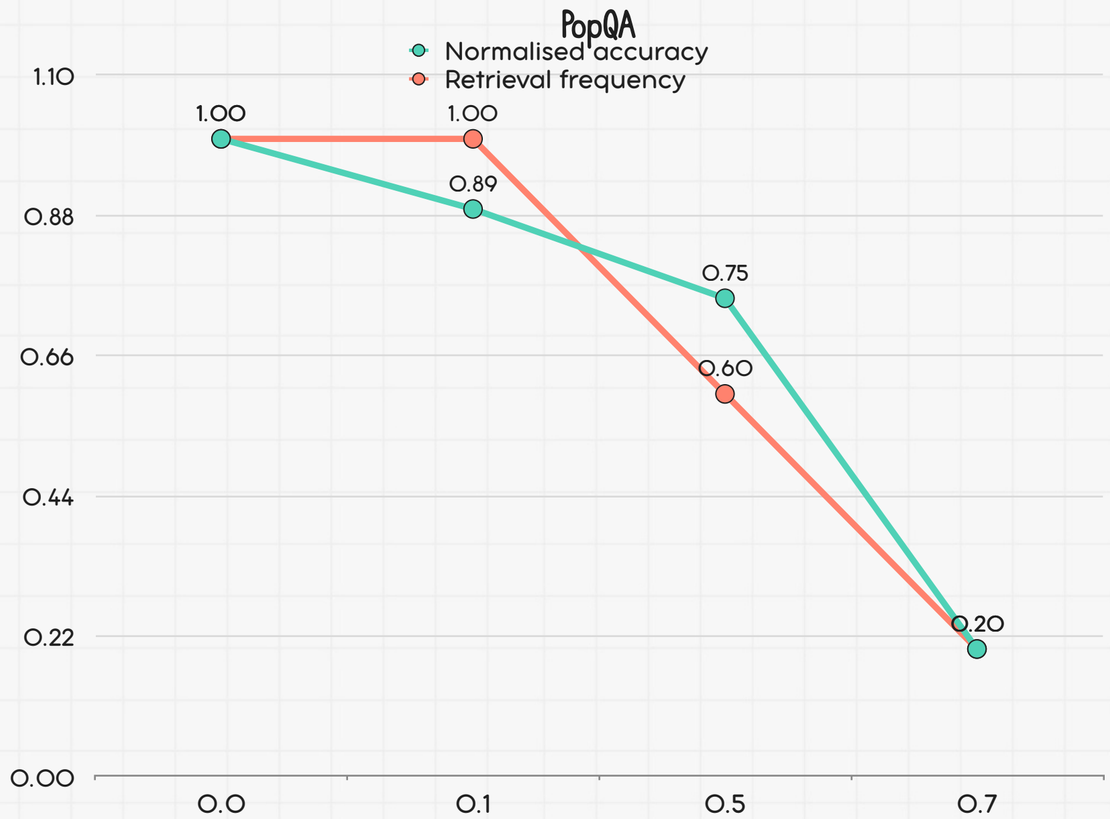




Raising w Sup favours well-supported, conservative output (precision up) at the cost of fluency (MAUVE down): chosen at inference, no retraining. Values read off the paper's figure (approximate).

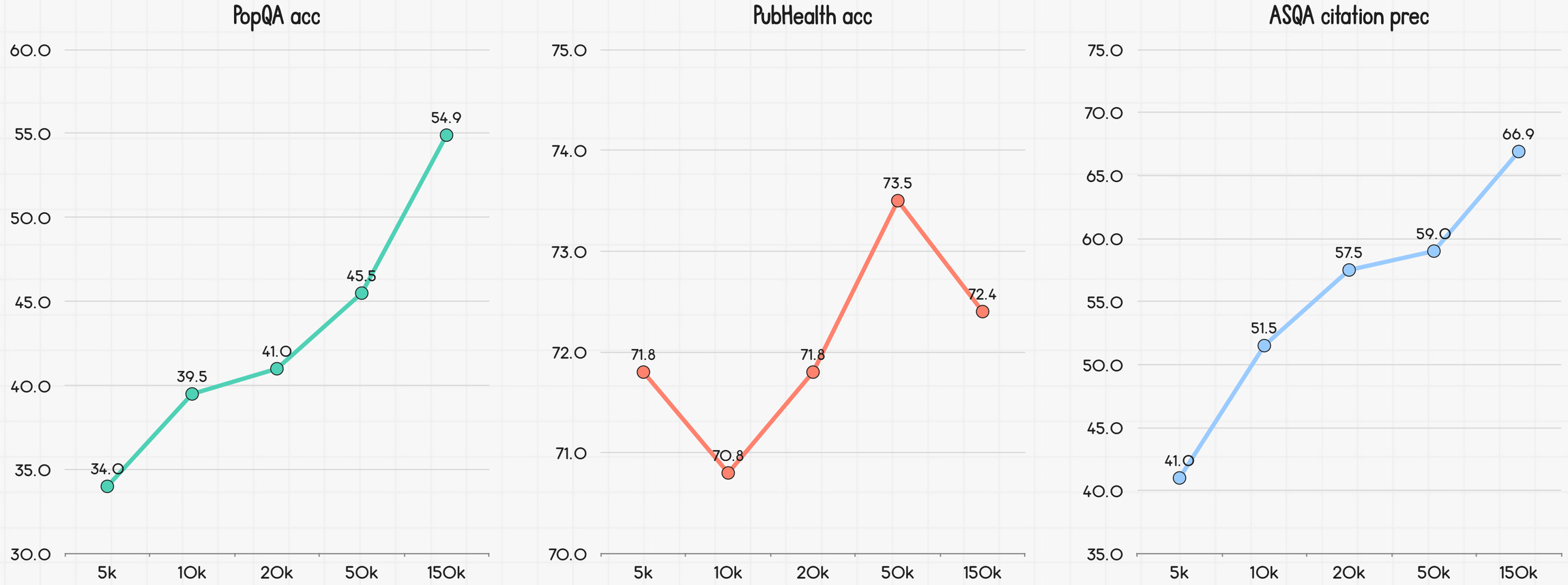
Efficiency vs accuracy: the threshold δ (Fig. 3c)





Larger δ means fewer retrievals. PubHealth barely loses accuracy when retrieval stops; PopQA collapses. Values read off the paper's figure (approximate).

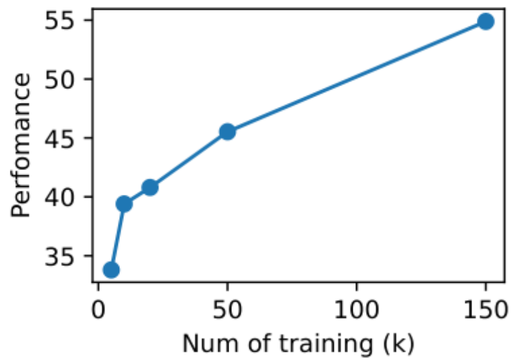
Training-data scale (Fig. 4a–c)



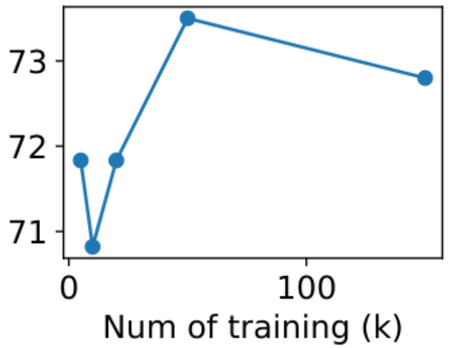
Self-RAG 7B fine-tuned on random subsets. PopQA and ASQA keep rising at 150k, so more data would likely help; Llama2-FT shows no comparable gain from 50k → 150k. Approximate values from the figure; 50k and 150k points from the ablation and Table 2.

From the paper: Figure 4

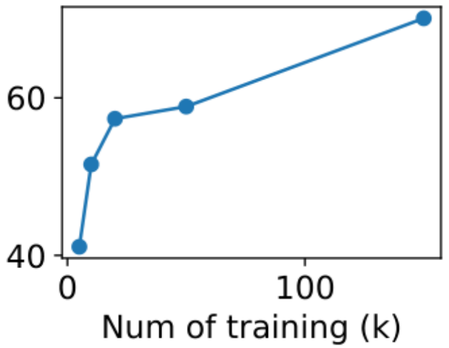
Asai et al. 2024, Fig. 4



(a) PopQA



(b) PubHealth



(c) ASQA (prec)

	Pop	Bio.
S & P	92.5	70.0
IsRel	95.0	90.0
IsSup	90.0	85.0

(d) Human evaluation on PopQA and Bio generation.

- (a)-(c): Self-RAG 7B trained on 5k, 10k, 20k, 50k, 150k examples: PopQA accuracy, PubHealth accuracy, ASQA citation precision.
- (d): human evaluation on 50 samples each: S&P and correctness of the IsRel / IsSup predictions.

Human evaluation (Fig. 4d, App. Table 6)

50 samples each	PopQA	Bio
S&P: supported and plausible	92.5	70.0
IsRel judged correct	95.0	90.0
IsSup judged correct	90.0	85.0

S&P (Menick 2022): plausible = on-topic, reasonable answer; supported = evidence suffices to verify it. Cases where Self-RAG itself predicted Irrelevant / No support are excluded. Reflection tokens are mostly meaningful, not decorative.

Failure: 'Who is the author of The Lie?'

- Output: 'The author of The Lie is Sam Harris' [IsRel=Relevant] [IsSup=Fully]
- Passage is about Harris's book 'Lying', a different title
- Both reflection tokens are wrong: plausible but not supported

Partial: 'Tell me a bio about Sergei Bodrov'

- First sentence fully supported by the passage
- Later 'Continue' sentences (born in Moscow; Guild president) are not in the evidence
- Plausible, only partially supported

Qualitative examples (App. Table 7, Self-RAG 13B)

Sunday as the 7th day (ASQA)

- Two retrievals, both Relevant and Fully supported
- Cites Constantine's decree of AD 321 from two passages
- Shows multi-segment, multi-citation answers

Rory Tapner bio

- Claims CEO of Coutts 'since 2010'
- Passage: stepped down in 2015
- Model emits IsSup = Contradictory: the error is flagged, enabling hard filtering

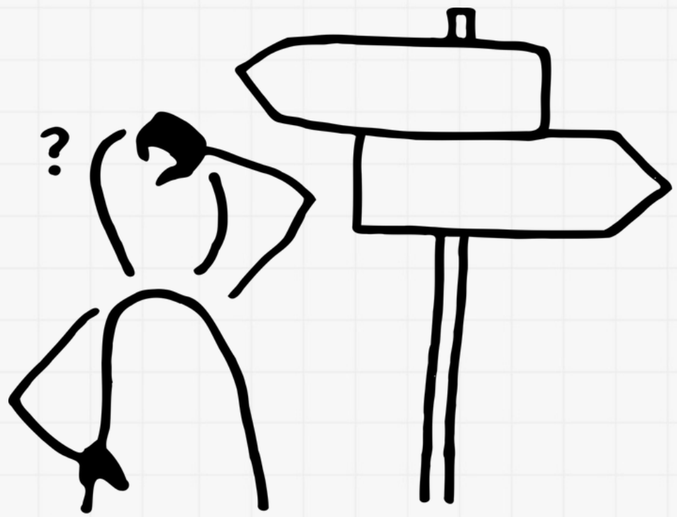
G. Venugopal bio

- 'Sung over 300 songs in Malayalam, Tamil and Hindi'
- Passage supports 300+ films, not the languages
- Model emits Partially supported: a fine-grained, honest label

05

Critical analysis

- Strengths
- Limitations: supervision, method, evaluation
- Is the evidence sufficient for the claims?
- Assumptions and how to improve



Strengths

Simple mechanism

Vocabulary expansion + SFT + reranking. No RL, no extra model at inference.

Controllable

δ and w trade retrieval cost, precision and fluency at test time.

Verifiable

Per-segment citations with mostly correct self-labels (85–95% human agreement).

Strong controls

Llama2-FT isolates data from framework; training and inference ablations.

Broad evaluation

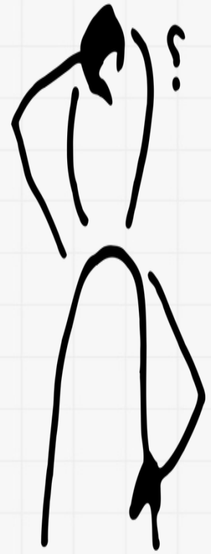
Six tasks, three formats; correctness, factuality, fluency and citations.

Open release

Code, models and data released; reproducible pipeline.

Limitations (1/3): supervision

- **All labels trace back to GPT-4:** temperature 1, a single sample per item; any GPT-4 bias (e.g. towards verbose answers being 'useful') is distilled twice.
- **Tiny human validation:** 20 items per token type checked by the authors themselves.
- **Error cascade:** the critic agrees with GPT-4 only 80.2% on IsRel and 73.5% on IsUse; its mistakes become training targets for M.
- **Agreement is with GPT-4, not humans:** critic accuracy measures imitation of the teacher.
- **Evidence of propagation:** 'The Lie' example: Relevant + Fully supported on the wrong book.



Limitations (2/3): method

- **Frozen retriever:** if the right passage is not in the top-K, reflection can only detect failure (IsRel = Irrelevant), not fix it; no query rewriting or re-retrieval loop.
- **One passage per segment:** a sentence that needs two passages, or a claim spanning sentences, is handled poorly; no multi-hop synthesis within a segment.
- **Heuristic scoring:** Eq. 3 adds a probability ([0,1]) to expected scores on different ranges ([0,1] and [−1,1]); the 0.5 credit, the utility scale and the default weights are hand-picked.
- **Little sensitivity analysis:** only w_{Sup} and δ are swept; w_{Rel} and w_{Use} are never ablated.
- **Inference cost unreported:** up to $B \cdot K = 10$ drafts per retrieval step plus prefills; no latency or throughput numbers.
- **Self-assessment is not verification:** IsSup is the model grading itself; calibration of critique probabilities is not studied.

Limitations (3/3): evaluation

- **Retrieval sources differ by task:** Self-RAG adds Google search results for Bio and open QA; shared retrieval is stated explicitly only for ASQA.
- **Copied baselines:** SAIL, Toolformer and CoVE numbers come from their papers, with different setups.
- **Gameable metrics:** match accuracy rewards long outputs; FactScore rewards short ones (the paper's own 7B > 13B note).
- **Ablation scale:** 50k-example models, 150 ASQA samples, no seeds or variance.
- **Human evaluation:** 50 examples per dataset; excludes cases the model flagged as irrelevant or unsupported.
- **Headline wording:** 'outperforms ChatGPT' holds on 4 of 6 tasks, not all; recall and str-em on ASQA remain below ChatGPT.

Is the evidence sufficient for the claims?

Claim	Evidence	Verdict
Beats open LLMs and standard RAG	All six tasks, same retriever, Llama2-FT control	Strong
Gains come from the framework, not data	Llama2-FT with identical 150k data; no-critic ablation	Strong
Outperforms ChatGPT	Wins 4 of 6 tasks; loses TQA, ARC, ASQA recall / str-em	Partial
Improves factuality and citations	FactScore, citation precision, 2% vs 15-20% grounding analysis	Strong
Controllable without retraining	w Sup sweep (Fig. 3b), δ sweep (Fig. 3c)	Moderate (one weight swept)
Reflection tokens are reliable	Critic vs GPT-4 74-94%; humans 85-95% on 50 samples	Moderate
Efficient	No latency measurements	Not shown

Assumptions and how to improve

Hidden assumptions

- A good answer can be built sentence by sentence
- Each sentence can be supported by a single passage
- GPT-4 judgments of relevance, support and utility are a valid proxy for humans
- Wikipedia (2018/2020) + web snippets contain the needed facts
- Critique-token probabilities are calibrated enough to add linearly

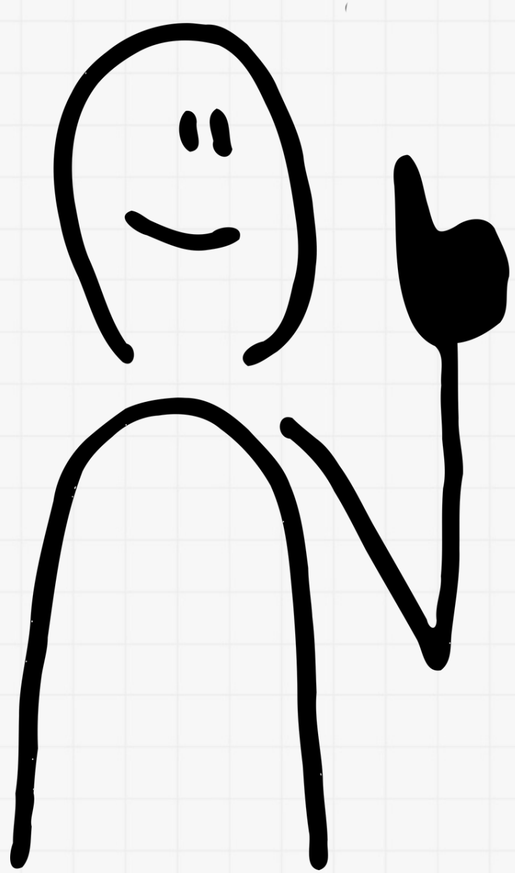
Improvements

- Re-retrieve or rewrite the query when every IsRel is low (later done by CRAG)
- Train or fine-tune the retriever jointly
- Learn or calibrate w on a small dev set (temperature scaling)
- Let a segment condition on a small set of passages chosen by IsRel
- Preference optimisation (DPO) on critique-ranked pairs, so M also improves its outputs
- Report latency; prune low-IsRel branches before generating

06

Impact

- From always-on RAG to retrieval on demand
- Follow-up research
- Connections to modern LLM systems



Timeline

2020 → **2023** → **2024** → **2024** → **2025**

RAG	Self-RAG	CRAG, Adaptive-RAG	Agentic RAG	Search as a tool
Lewis et al.: retrieval as a latent variable, always on	Retrieval on demand + self-critique tokens (ICLR 2024)	Grade documents, fall back to web search; route queries by difficulty	LangGraph / LlamaIndex recipes: retrieval, hallucination and answer 'graders'	Assistants decide when to search and cite; RL-trained search agents

Lasting idea: special tokens as an interface for control and self-evaluation, and test-time reranking instead of retraining.

Follow-up work building on Self-RAG

Work	What it does	Relation to Self-RAG
CRAG (Yan et al. 2024)	Lightweight evaluator grades retrieved documents; triggers web search or knowledge refinement	Fixes the frozen-retriever gap (detect and correct)
Adaptive-RAG (Jeong et al. 2024)	Classifier routes queries: no retrieval / single-step / multi-step	Generalises 'retrieve only when needed'
RankRAG (Yu et al. 2024)	One LLM instruction-tuned to rank contexts and generate	Same single-model judge + writer philosophy
Search-R1 (Jin et al. 2025)	RL teaches an LLM to interleave reasoning and search calls	Replaces offline labels with outcome rewards
LangGraph / LlamaIndex recipes	Prompted graders as nodes in an agent graph	IsRel / IsSup / IsUse as prompts instead of tokens

Self-RAG and modern LLM systems

Retrieve token → tool calls	Assistants decide when to call search, exactly the role of Retrieve, generalised to any tool.
IsSup → cited answers	Citation-grounded answers and 'is this claim supported?' checks in production search assistants.
Critique → LLM-as-judge	Graders for relevance, hallucination and helpfulness in RAG pipelines and evaluation.
Reranking → test-time compute	Best-of-N and verifier-guided search spend inference compute to raise quality without retraining.
Offline labels → RL with outcomes	Reasoning models trained with RL learn when to search from final-answer rewards.

Key takeaways

1. Plain RAG retrieves indiscriminately and never checks its output.
2. Self-RAG trains one LM to emit reflection tokens: retrieve on demand, judge relevance, verify support, rate utility.
3. Critique is learned offline (GPT-4 → critic → corpus) and trained with plain next-token loss: no RL.
4. At inference, critique probabilities form a tunable score for segment-level beam search: controllable without retraining.
5. Best open model on six tasks, large citation gains; the Llama2-FT control shows the framework, not the data, drives it.
6. Limits: GPT-4-derived supervision, frozen retriever, heuristic scoring, unreported cost.

Team Contribution



Student	Contribution	Score	Main work
[Student A]	[Equal / Partial]	[1.0]	[e.g. background & problem]
[Student B]	[Equal / Partial]	[1.0]	[e.g. method & equations]
[Student C]	[Equal / Partial]	[1.0]	[e.g. experiments & analysis]
[Student D]	[Equal / Partial]	[1.0]	[e.g. critique, impact, references]

Scores reflect actual relative contribution (preparation, understanding, slides, presentation), as the rubric requires.

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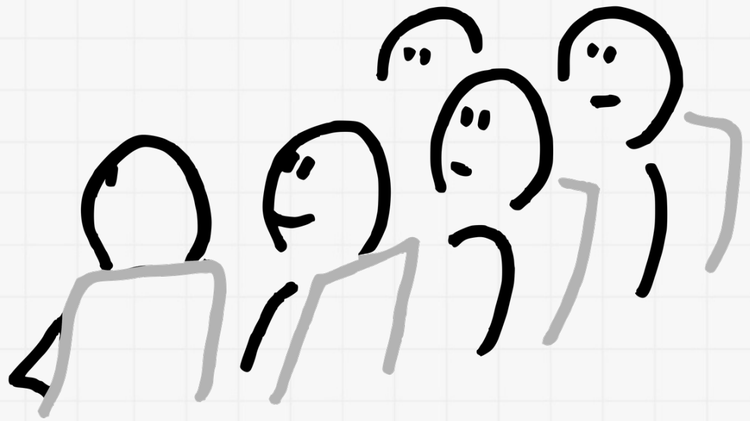
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Thank you!

Questions?

Paper: arxiv.org/abs/2310.11511
Code and models: selfrag.github.io

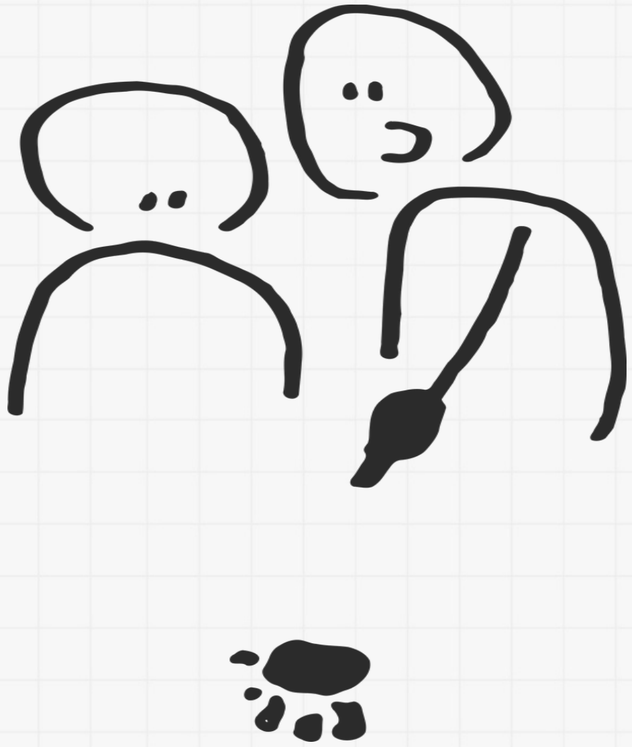




B

Backup slides

- Q&A bank with short answers
- GPT-4 prompt scales for IsSup and IsUse
- Evaluation instructions



Q&A bank (1/2)

Why not always retrieve?

Hurts versatility and adds distractors: Ret-ChatGPT PubHealth 70.1 → 54.7; never-retrieve costs PopQA 20.8 but PubHealth 0.5.

Why mask retrieved passages?

They are inputs from R; predicting them teaches memorisation or invented passages. They are still attended to.

Why a separate critic?

Labelling 150k examples × sentences × passages × aspects with GPT-4 is costly and not reproducible; a 7B critic reaches 74–94% agreement.

Why not PPO?

About 4 models in memory, on-policy sampling, unstable; and behaviour is then fixed. SFT + test-time weights is cheaper and tunable.

What does Eq. 3/4 mean?

Segment probability + weighted expected critique values; the grounded candidate beats the fluent one (2.585 vs 1.31).

Why δ = 0.2?

Biases towards retrieval on factual tasks; argmax (δ about 0.5) retrieves too rarely: PopQA 45.5 → 28.3.

Q&A bank (2/2)

Is the gain from data?

No: Llama2-FT with the same 150k data trails (PubHealth 64.3 vs 72.4; citation precision 5.0 vs 66.9).

Why does 7B beat 13B on Bio?

FactScore measures precision; 7B writes shorter, precisely grounded outputs.

Biggest limitation?

GPT-4-derived labels distilled twice with tiny human checks; or the frozen retriever; or unreported B·K cost.

How would you improve it?

Re-retrieval when IsRel is low; learned / calibrated weights; multi-passage segments; DPO on critique pairs.

Why the 'Continue' value?

One passage can support several sentences (biographies); avoids redundant retrieval calls.

Why weight IsUse by 0.5?

It ignores factuality and has the lowest critic accuracy (73.5%); down-weighting stops fluent but unsupported answers winning.

GPT-4 prompt scales (App. D, Tables 11–12)

IsSup: entailment scale

- Fully supported: all information in the output is supported by the evidence (almost identical to part of it)
- Partially supported: supported to some extent, but major information is not in the evidence
- No support / Contradictory: ignores, is unrelated to, or contradicts the evidence
- Judge only against the evidence; no external knowledge

IsUse: perceived utility

- 5: complete, highly detailed, fully satisfies the need
- 4: mostly fulfils it; minor improvements possible
- 3: acceptable; major additions needed
- 2: addresses the main request but incomplete or not relevant
- 1: barely on-topic or irrelevant
- Example: 'UK PM as of 2023?' → 'Boris Johnson was PM 2019–2022' rated 2

Evaluation instructions (App. Table 5)

Dataset	Zero-shot instruction (paraphrased)
ARC-Challenge	Given four answer candidates A–D, choose the best; answer with the capital letter only
PubHealth	Is the statement correct? Say 'true' or 'false', lower-case, no period
Bio generation	Tell me a bio about [Person Name]
ASQA (baselines)	ALCE prompt: answer using only the search results, cite 1–3 documents per sentence with [1][2]…
ASQA (Self-RAG)	Answer the question; it may be ambiguous with several correct answers, so give a long answer covering all
PopQA, TriviaQA	No instruction; the question itself is the input